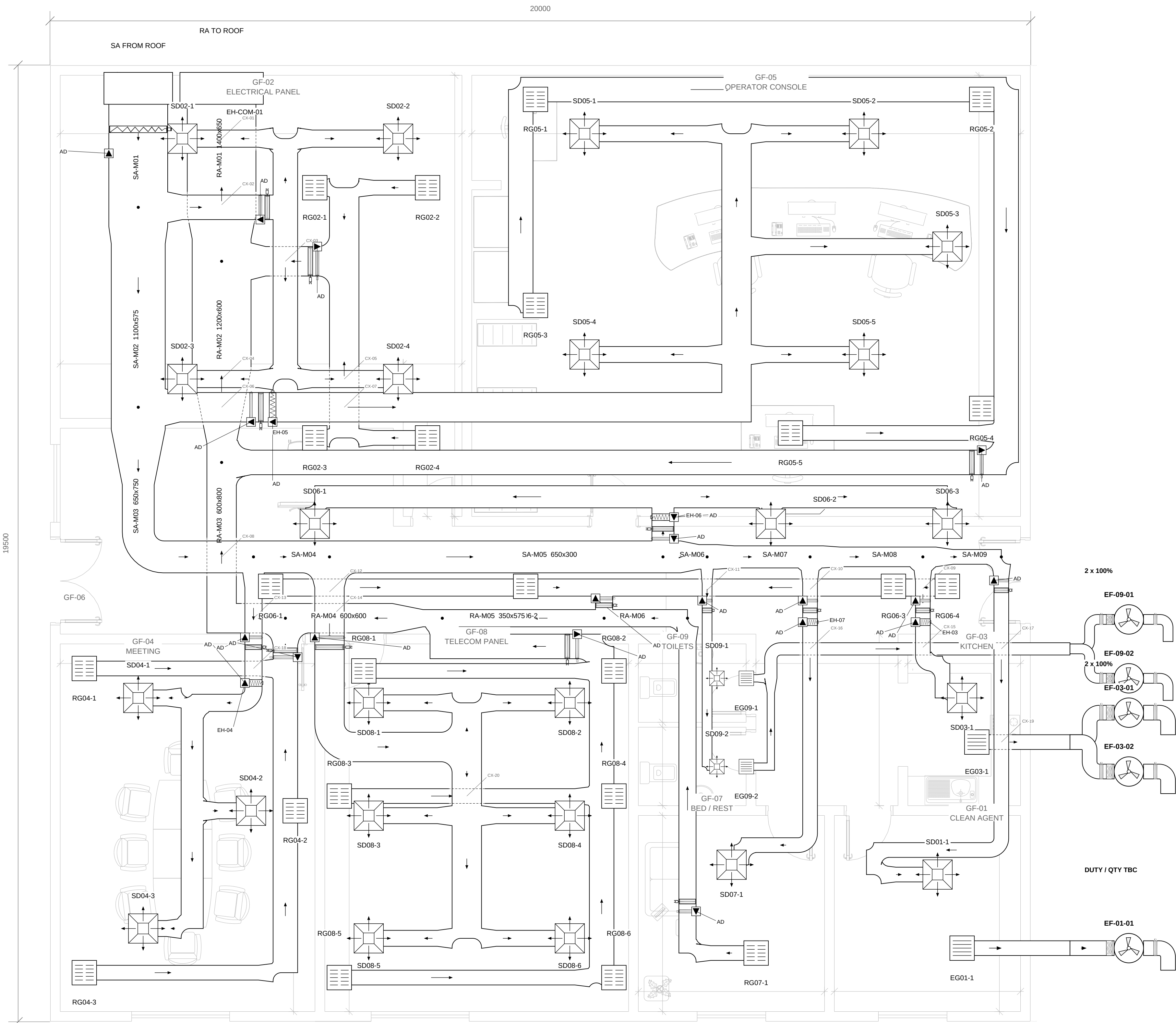




ROOM AIRFLOW SCHEDULE

GF	Room	SA L/s	RA L/s*	EA L/s*
02	ELECTRICAL PANEL	664.8	605.5	-
05	OPERATOR CONSOLE	777.7	718.4	-
04	MEETING	449.2	389.9	-
08	TELECOM PANEL	884.6	825.3	-
06	CORRIDOR	418.2	358.9	-
07	BED / REST	168.2	108.9	-
09	TOILETS	46.2	-	64.9
03	KITCHEN	147.6	-	147.6
01	CLEAN AGENT	141.3	-	TBC

DUCT / SYMBOL LEGEND

- SA - supply duct
- RA - return duct
- EA - dedicated extract
- SD - four-way supply diffuser
- RG - ducted return grille
- EG - extract grille
- VCD - balancing damper
- MFD* - conditional fire damper
- RCD - return control damper
- EH - inline electric heater
- AD - access door

Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

CX tags identify separate-service crossings: dashed lower duct walls, continuous upper walls. See D-020 for the crossing register and vertical-clearance requirements. Levels remain TBC.

* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification.

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-001

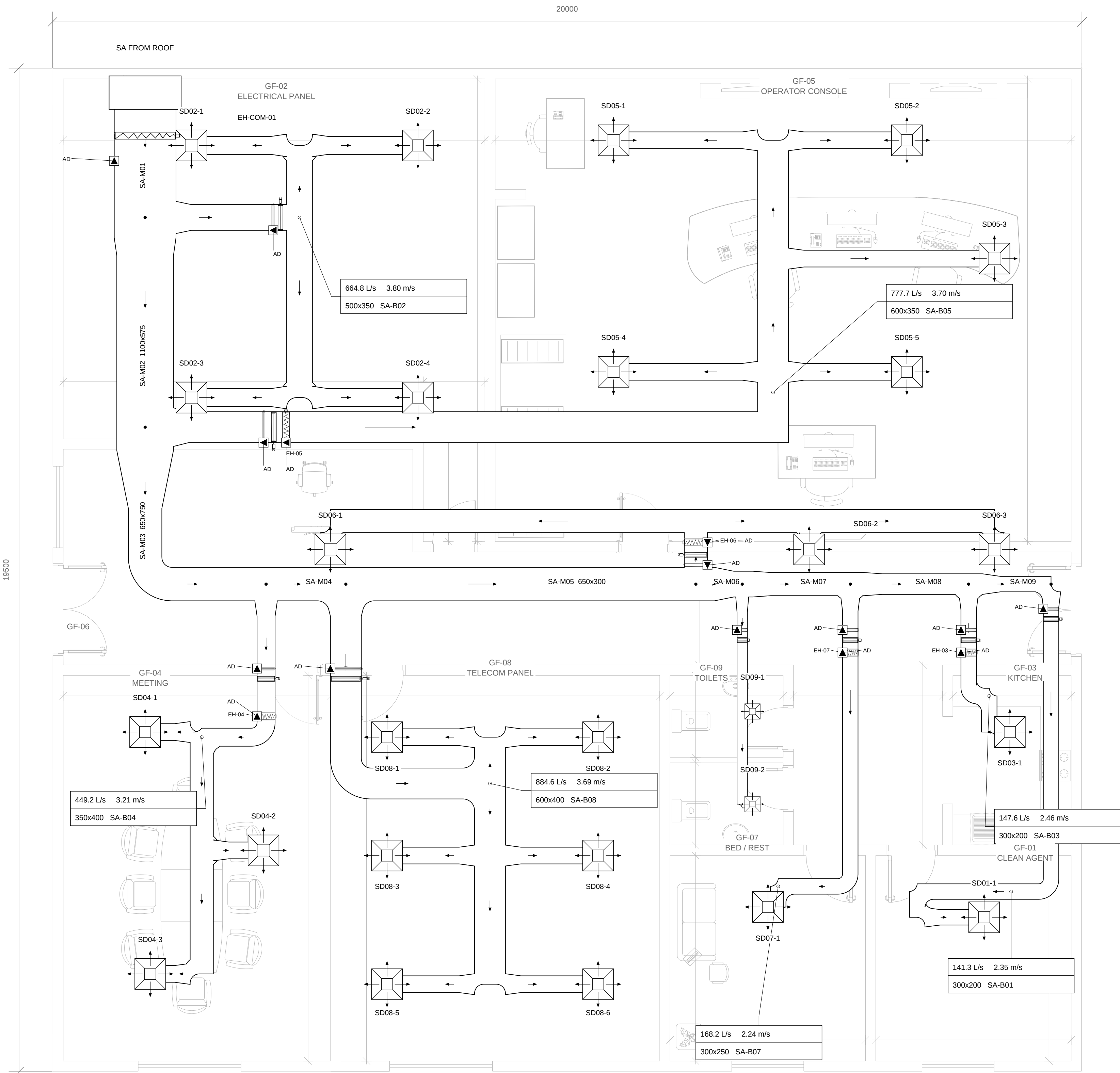
SCALE: 1:50 AT A1

REVISION

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SHEET 1 OF 21



SUPPLY MAIN SECTIONS

Section	W x H mm	L/s	m/s
SA-M01	1200x650	3697.8	4.74
SA-M02	1100x575	3033.0	4.80
SA-M03	650x750	2255.3	4.63
SA-M04	650x575	1806.1	4.83
SA-M05	650x300	921.5	4.73
SA-M06	500x250	503.3	4.03
SA-M07	450x250	457.1	4.06
SA-M08	400x200	288.9	3.61
SA-M09	300x200	141.3	2.35

ROOM SUPPLY BRANCHES

GF	W x H mm	L/s	m/s
02	500x350	664.8	3.80
05	600x350	777.7	3.70
04	350x400	449.2	3.21
08	600x400	884.6	3.69
06	450x300	418.2	3.10
07	300x250	168.2	2.24
09	200x150	46.2	1.54
03	300x200	147.6	2.46
01	300x200	141.3	2.35

Room headers retain their branch size. Diffuser tags identify individual outlets. Round runout / neck diameters and each terminal airflow are listed on D-004 to D-006 and D-010.

MFD* is a conditional rated-penetration provision. VCD-Snn and FSD-Snn correspond to the GF-rn supply branch. Inline heaters are upstream of that room's terminal take-offs.

Continuous main tapers, flared branch boots and terminal adaptors: D-011. AD tags identify maintenance access doors. Proposed taper lengths and device coordinates are in the registers.

Total HAP-reference supply: 3697.80 L/s. Three 50% PAUs: two duty, rotating available alternate. Boxed callouts show branch flow, velocity and clear size. No room sensors are overlaid on this duct plan.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SUPPLY AIR DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-002

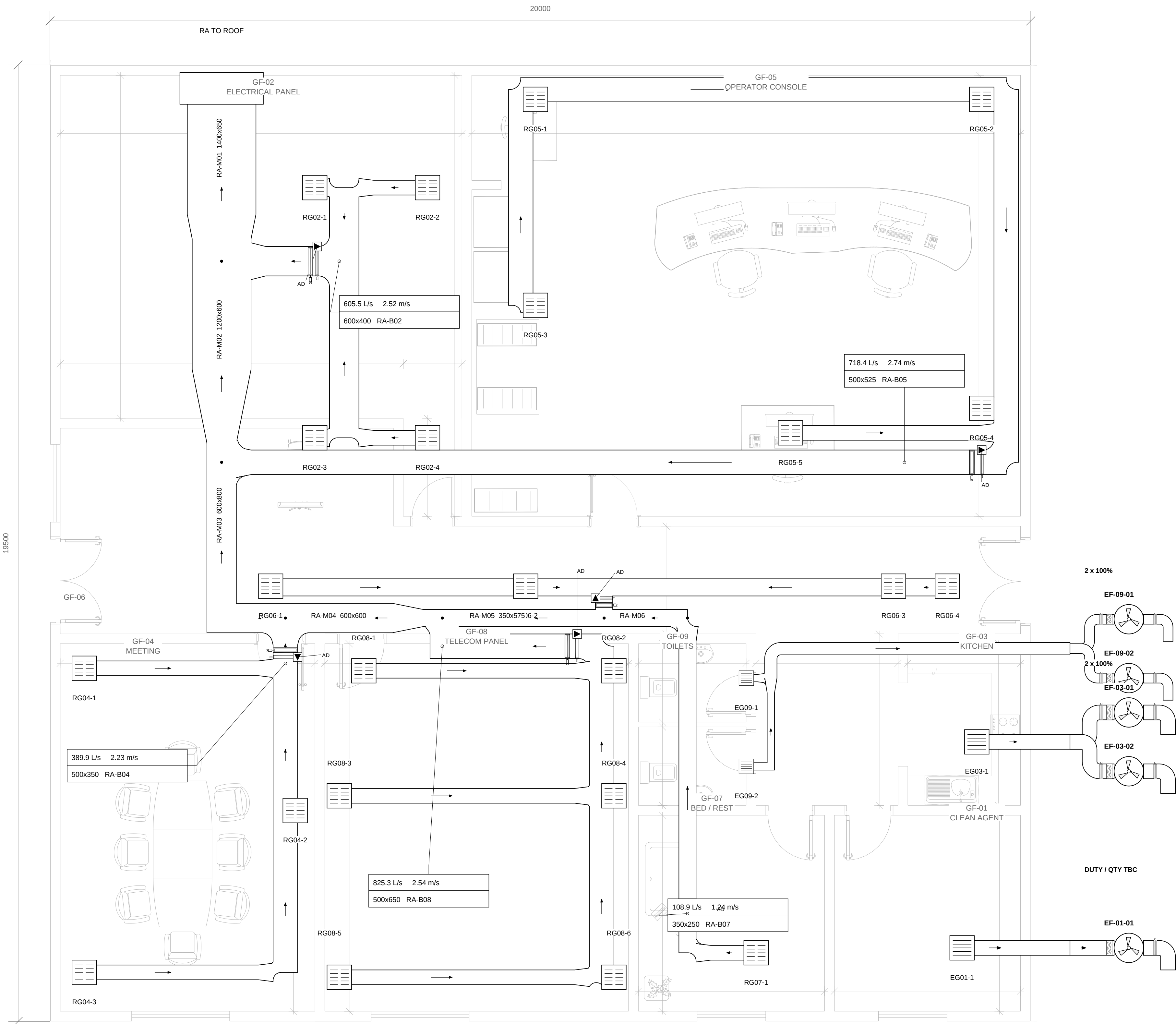
SCALE: 1:50 AT A1

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RETURN MAIN SECTIONS

Section	W x H mm	L/s	m/s
RA-M01	1400x650	3006.7	3.30
RA-M02	1200x600	2401.2	3.34
RA-M03	600x800	1682.9	3.51
RA-M04	600x600	1293.0	3.59
RA-M05	350x575	467.7	2.32
RA-M06	350x300	108.9	1.04

ROOM RETURN / EXTRACT

GF	Air	W x H mm	L/s*
02	RA	600x400	605.5
05	RA	500x525	718.4
04	RA	500x350	389.9
08	RA	500x650	825.3
06	RA	350x550	358.9
07	RA	350x250	108.9
09	EA	250x200	64.9
03	EA	300x200	147.6
01	EA	TBC	TBC

No central return: GF-09 toilets, GF-03 kitchen and GF-01 clean-agent. Kitchen / toilet fans: 2 x 100% with duty rotation. CAG duty and quantity remain unverified.

Extract fan symbols are schematic equipment markers; actual fan and louver dimensions depend on manufacturer selection. Wall outlets face downwards.

* Return total 3006.68 L/s is an assumed starting balance. Existing extract references do not verify positive pressure or outdoor-air compliance. Confirmed fire shuts down all HVAC; automatic fire extraction is not enabled.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

RETURN AIR AND EXTRACT DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-003

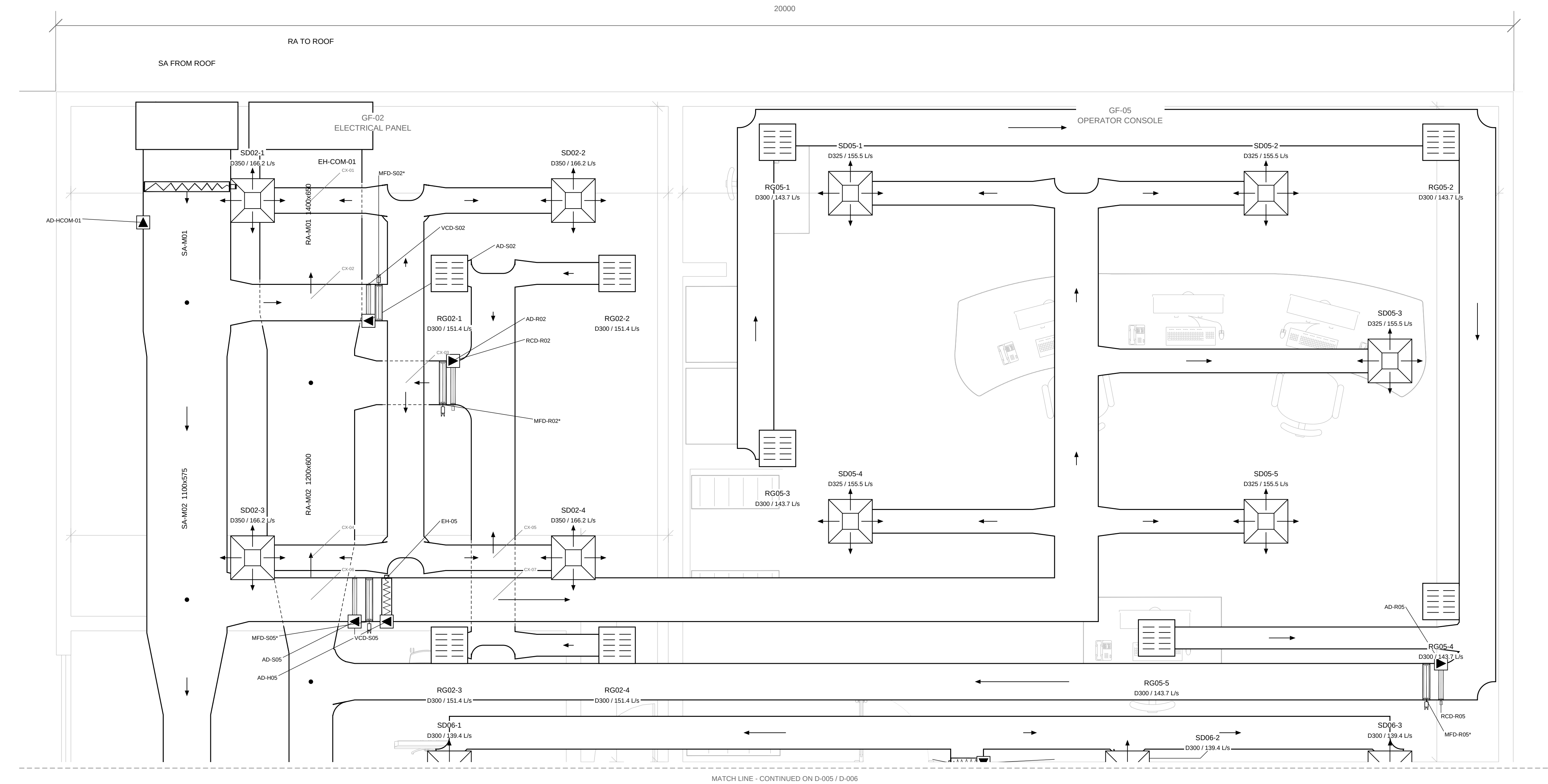
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REVISION

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SHEET 3 OF 21



NORTH-ROOM TERMINALS / BRANCHES

GF-02 ELECTRICAL PANEL

SA-B02 500x350 mm | 664.80 L/s | 3.80 m/s
RA-B02 600x400 mm | 605.46 L/s | 2.52 m/s
SD: 4 x 166.20 L/s; neck D350; face 600x600
RG: 4 x 151.37 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-05 OPERATOR CONSOLE

SA-B05 600x350 mm | 777.70 L/s | 3.70 m/s
RA-B05 500x525 mm | 718.36 L/s | 2.74 m/s
SD: 5 x 155.54 L/s; neck D325; face 600x600
RG: 5 x 143.67 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

Plan symbols identify VCD-S02 / S05, conditional MFD-S02 / S05*, RCD-R02 / R05 and conditional return MFDs. The component register retains the FSD tag prefix for the conditional motorized fire/smoke damper symbols.

Mains entering these rooms: SA-M01/M02/M03 and RA-M01/M02/M03. Source architectural orientation is retained. No structural grid, duct BOD, access envelope or manufacturer dimension is inferred from HAP room height.

Access doors: AD-Sn / AD-Rnn beside branch dampers; AD-Hnn beside heaters. Taper / access arrangements: D-011. Instrumentation: D-007 / D-012. Catalogue selection remains pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

NORTH ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-004

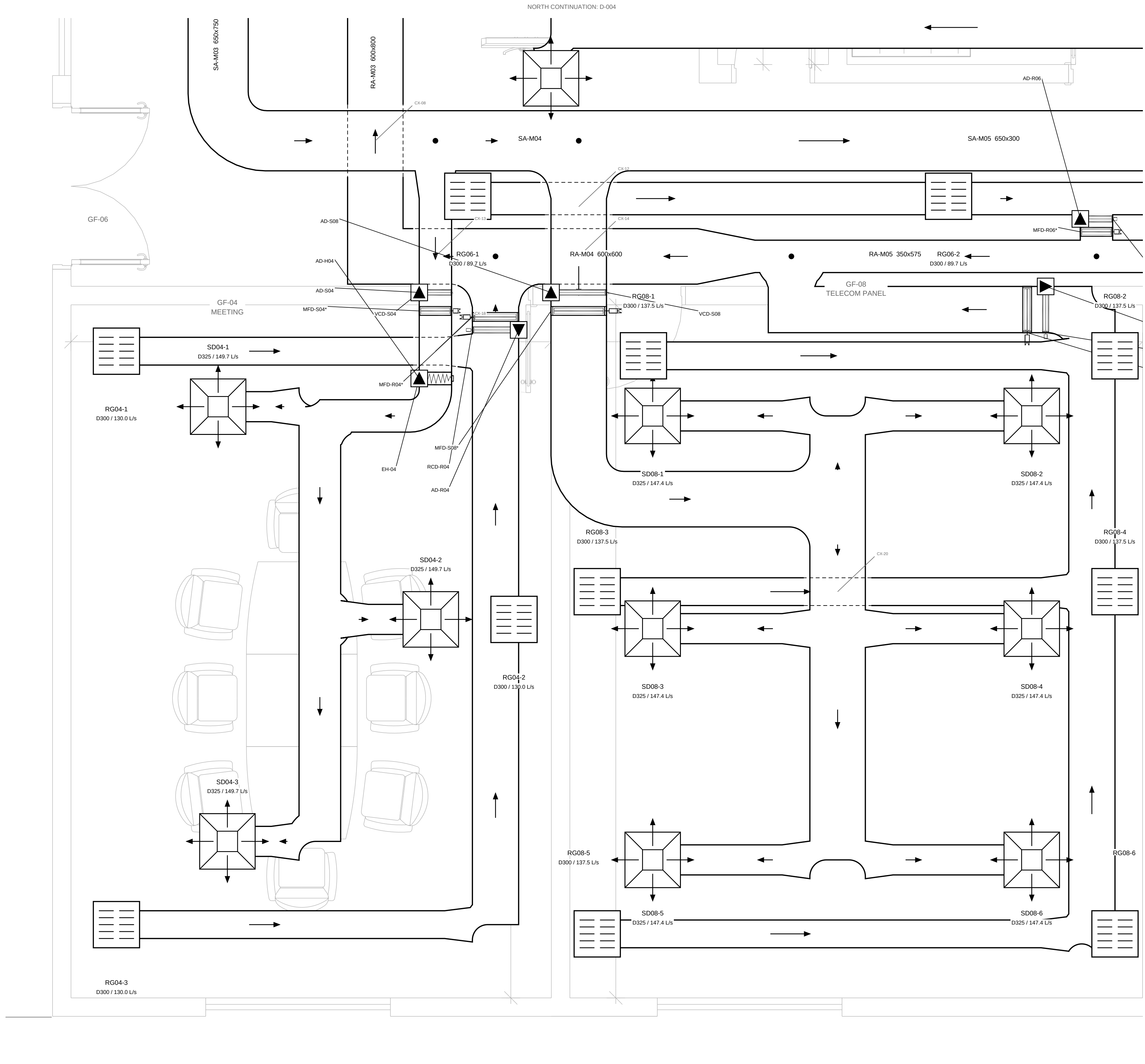
SCALE: 1:35 AT A1

REVISION

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ROOM BRANCH / TERMINAL DATA

GF-04 MEETING

SA-B04 350x400 mm | 449.20 L/s | 3.21 m/s

RA-B04 500x350 mm | 389.86 L/s | 2.23 m/s

SD: 3 x 149.73 L/s; neck D325; face 600x600

RG: 3 x 129.95 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-08 TELECOM PANEL

SA-B08 600x400 mm | 884.60 L/s | 3.69 m/s

RA-B08 500x350 mm | 825.26 L/s | 2.54 m/s

SD: 6 x 147.43 L/s; neck D325; face 600x600

RG: 6 x 137.54 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-06 CORRIDOR

SA-B06 450x300 mm | 418.20 L/s | 3.10 m/s

RA-B06 350x550 mm | 358.86 L/s | 1.86 m/s

SD: 3 x 139.40 L/s; neck D300; face 600x600

RG: 4 x 89.72 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

DUCT / SYMBOL LEGEND

- SA - supply duct
- RA - return duct
- EA - dedicated extract
- SD - four-way supply diffuser
- RG - ducted return grille
- EG - extract grille
- VCD - balancing damper
- MFD* - conditional fire damper
- RCD - return control damper
- EH - inline electric heater
- AD - access door

Red arrows show air direction; green outlines are supply and blue outlines are return. CX tags refer to D-020; dashed walls identify the lower duct. Same-service junctions are continuous open tees.

GF-04 meeting; GF-08 telecom; GF-06 corridor. Room targets +50 Pa relative to outdoors; actual pressure verification remains pending. Full flow / velocity / horizontal route data is listed on D-010.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-WEST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-005

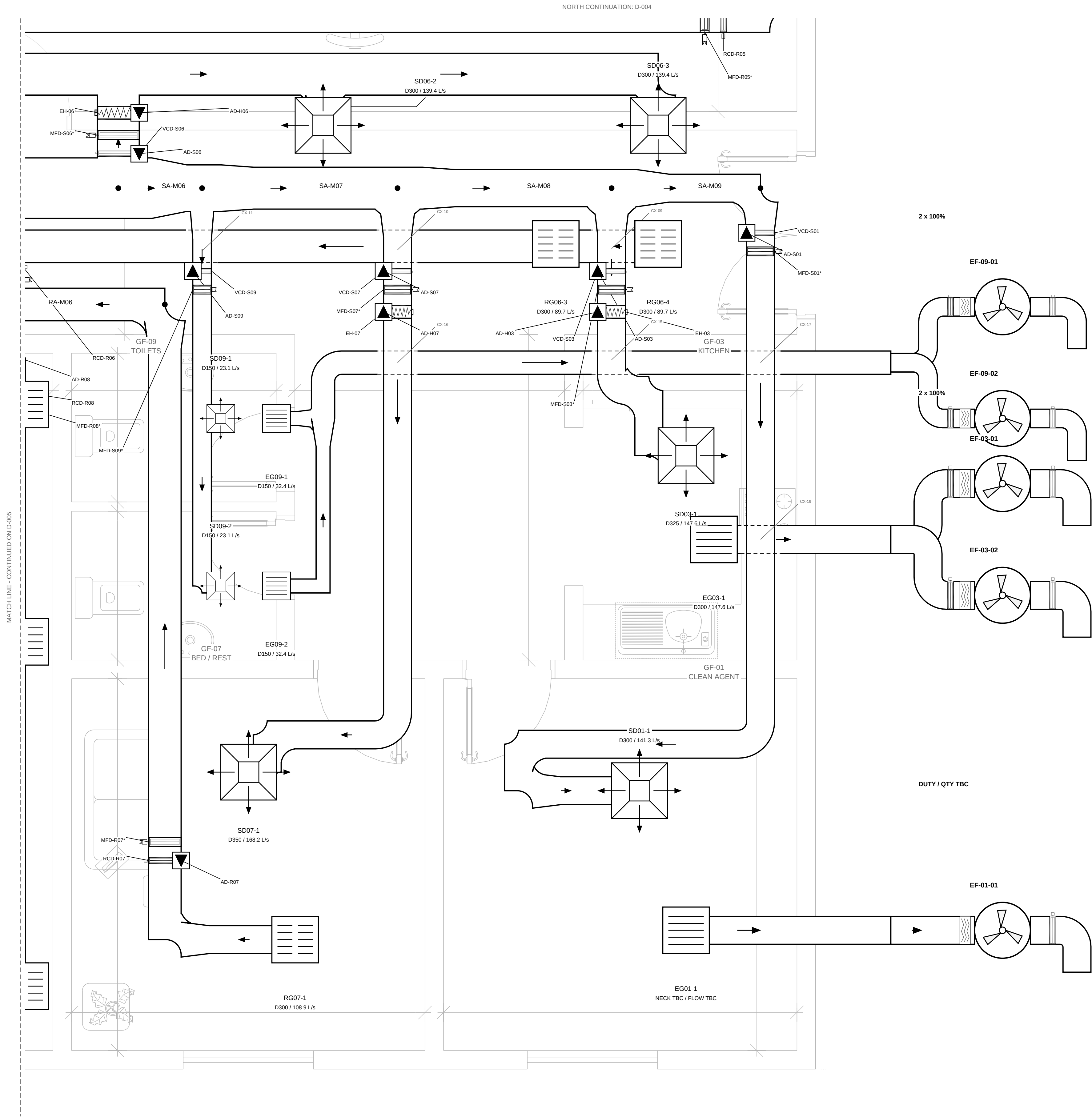
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CAG receives cooled PAU supply and has dedicated extract. No shared return from kitchen, toilet or CAG. All fan branches have motorized isolation; control / shutdown sequences are on the schematic set.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-EAST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-006

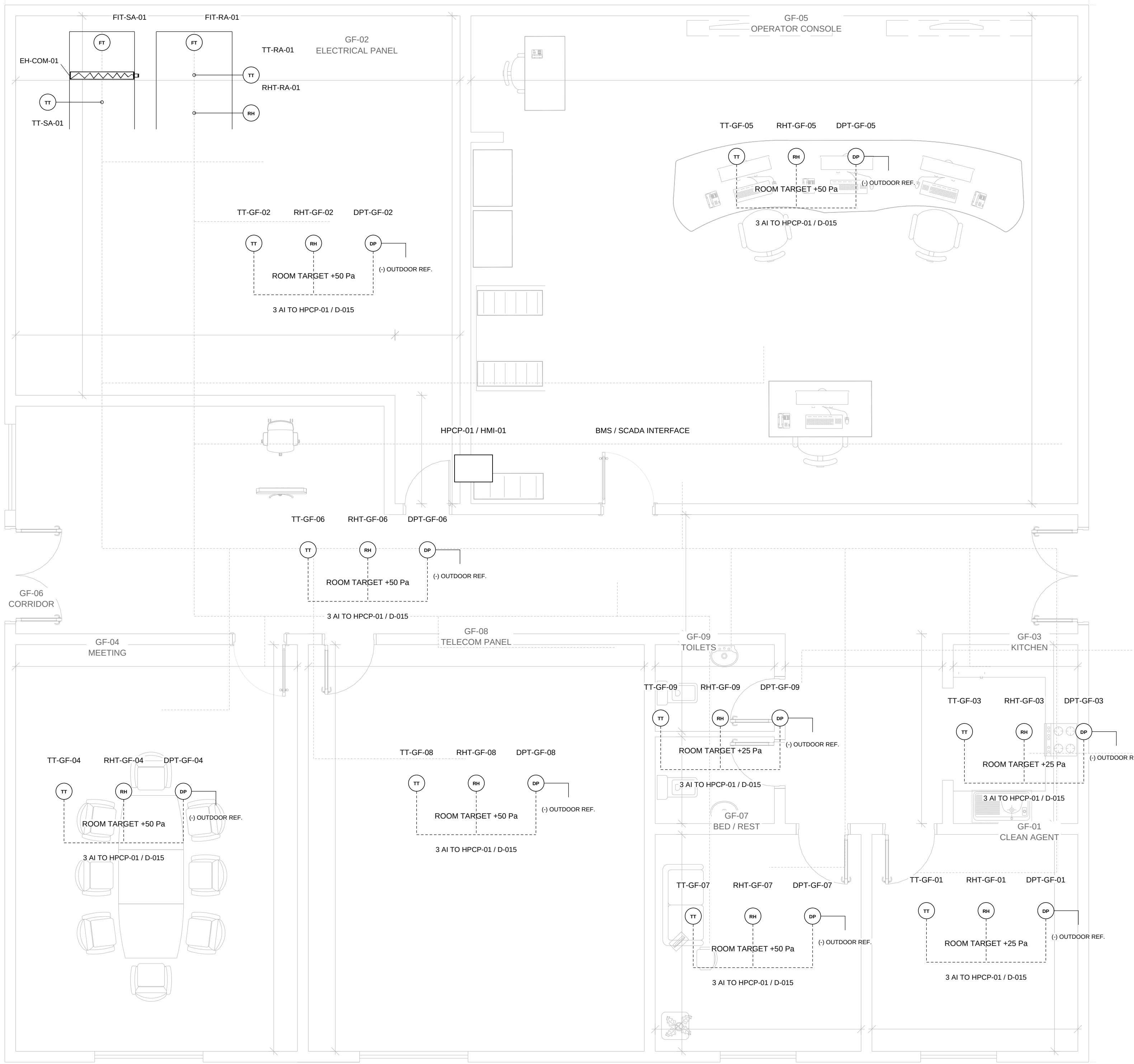
SCALE: 1:25 AT A1

REVISION

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ROOM FIELD INSTRUMENTS

GF	Temperature	Humidity	Pressure	Pa
02	TT-GF-02	RHT-GF-02	DPT-GF-02	50
05	TT-GF-05	RHT-GF-05	DPT-GF-05	50
04	TT-GF-04	RHT-GF-04	DPT-GF-04	50
08	TT-GF-08	RHT-GF-08	DPT-GF-08	50
06	TT-GF-06	RHT-GF-06	DPT-GF-06	50
07	TT-GF-07	RHT-GF-07	DPT-GF-07	50
09	TT-GF-09	RHT-GF-09	DPT-GF-09	25
03	TT-GF-03	RHT-GF-03	DPT-GF-03	25
01	TT-GF-01	RHT-GF-01	DPT-GF-01	25

MAIN DUCT / PLANT MONITORING

TT-SA-01 / TT-RA-01: main duct temperature

RHT-RA-01: main return humidity

FIT-SA-01 / FIT-RA-01: common airflow

Each PAU filter: PDT remote + PDI local

DPT-SF-A/B/C: PAU fan pressure proof

VS: PAU and ventilation-fan vibration

EM: feeders >7.5 kW, quantity to confirm

Signals report to HPCP-01 dedicated PLC and HMI-01. BMS / SCADA provides the specified monitoring, initial start and emergency-stop interfaces. Protocol, alarm limits, I/O module allocation and actuator fail positions remain pending.

DPT room references are outdoors, not the adjacent room. DPT-GF-06 is the corridor / building reference. Additional room DPTs and CAG humidity sensing are marked as added provisions in the register.

Room instrument locations are proposed. Plant/filter/fan instruments and heater AFS / TSHH protection: D-012. Final ranges, reference probes, mounting and routes require I&C coordination; operating sequence: S-004.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOM INSTRUMENTATION AND CONTROL LOCATIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-007

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REVISION

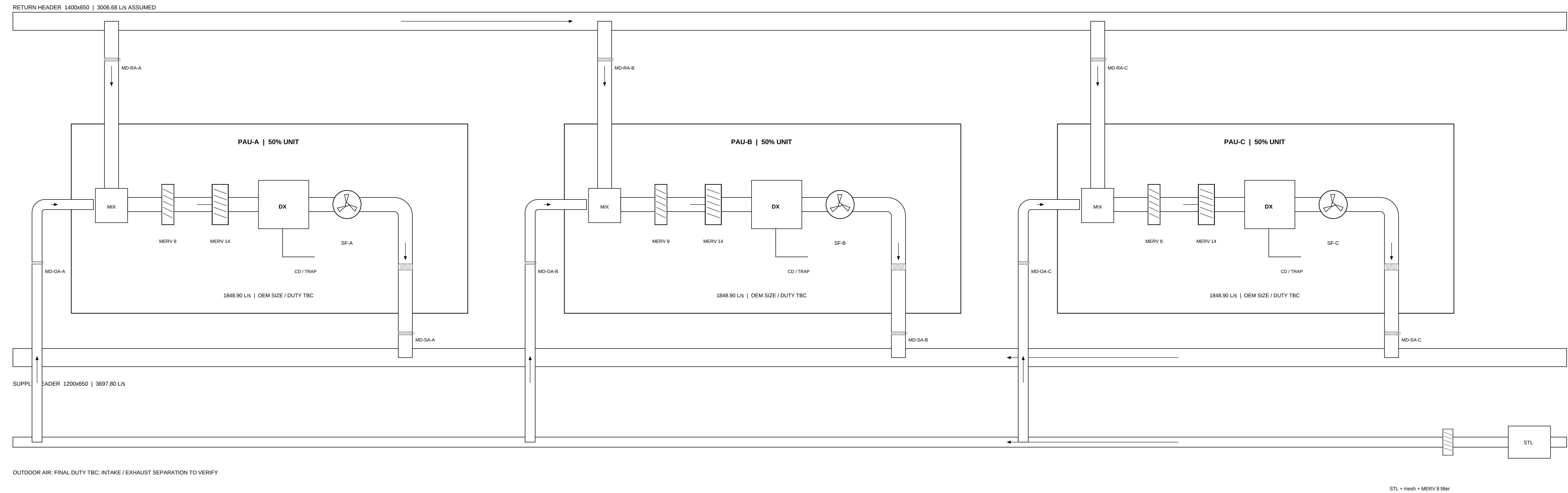
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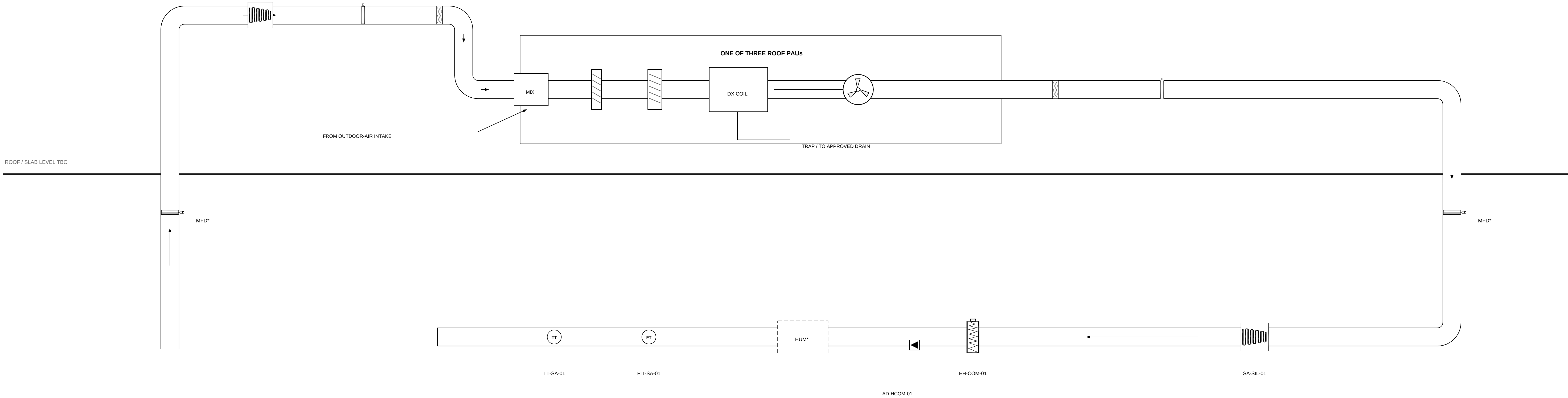
SHEET 7 OF 21

ADMINISTRATION BUILDING - PAU BANK AND INDOOR RISER SECTION

A ROOF CONNECTION ARRANGEMENT - SCHEMATIC FOOTPRINTS



B. ONE PAU / RISERS / INDOOR COMMON SUPPLY SECTION



All unit footprints / levels are schematic. MFD* requires approved compartmentation. Common heater and conditional humidifier are indoors. Filter DP, fan pressure / vibration and heater proof instruments: D-012 / S-001.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PAU BANK AND INDOOR RISER SECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

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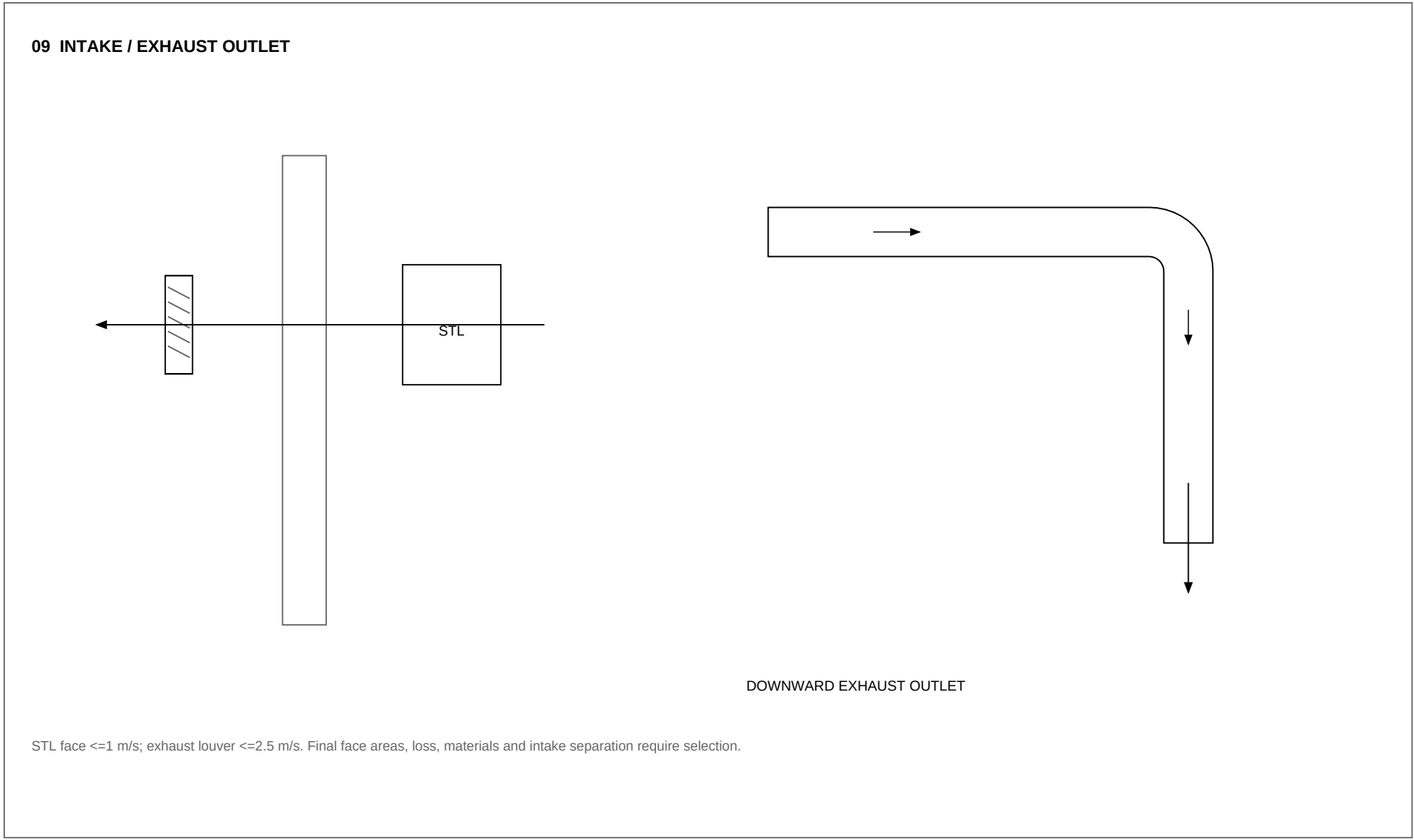
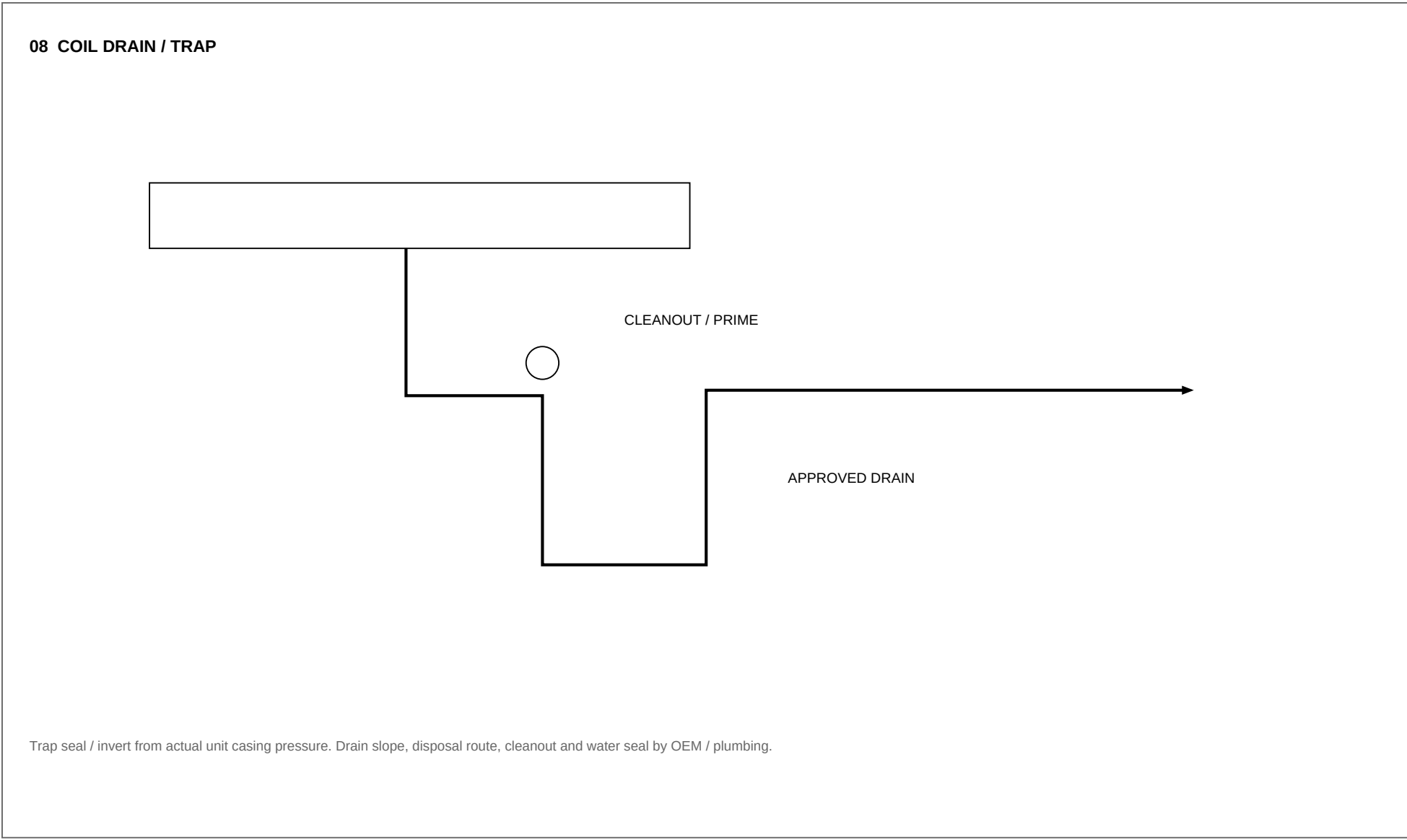
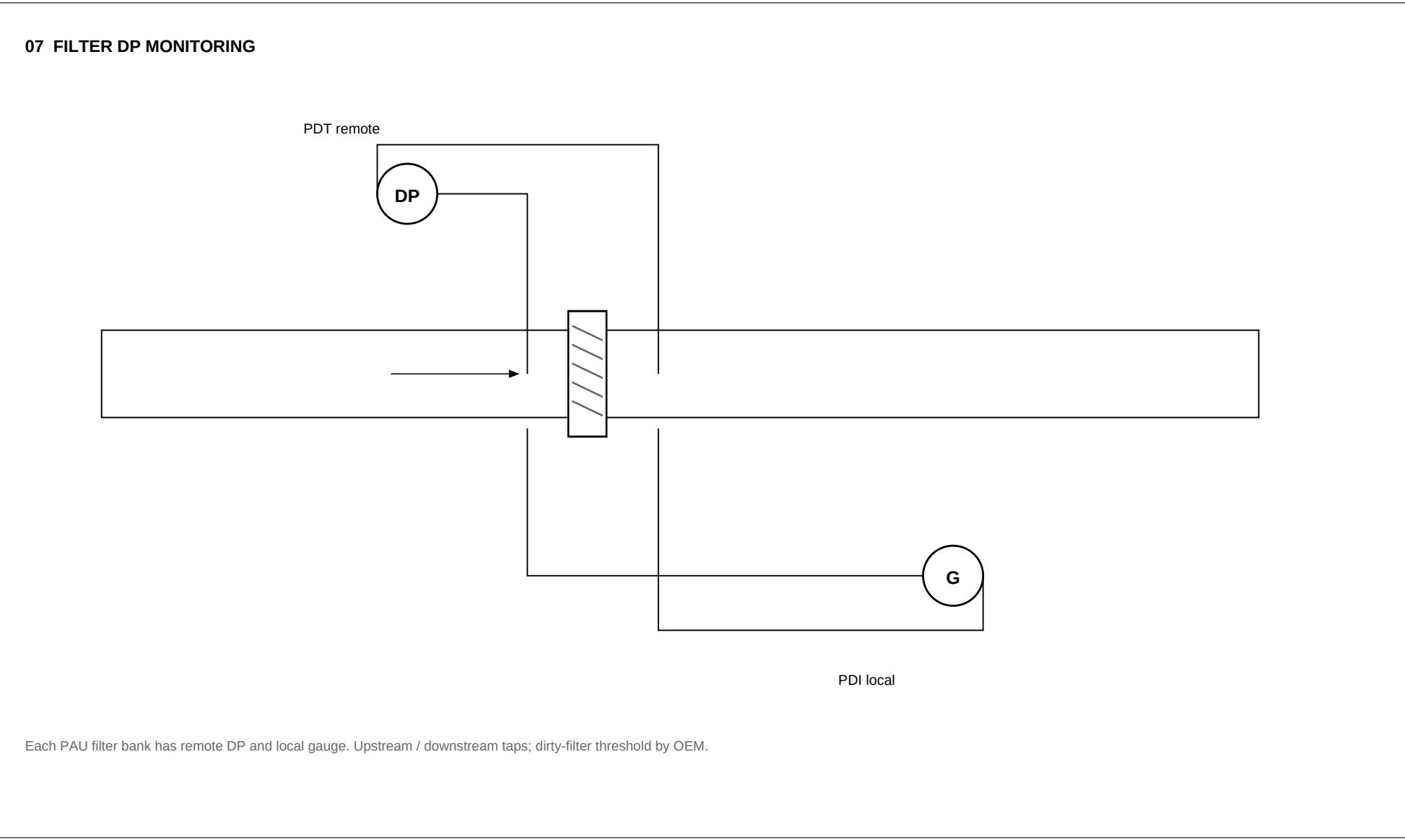
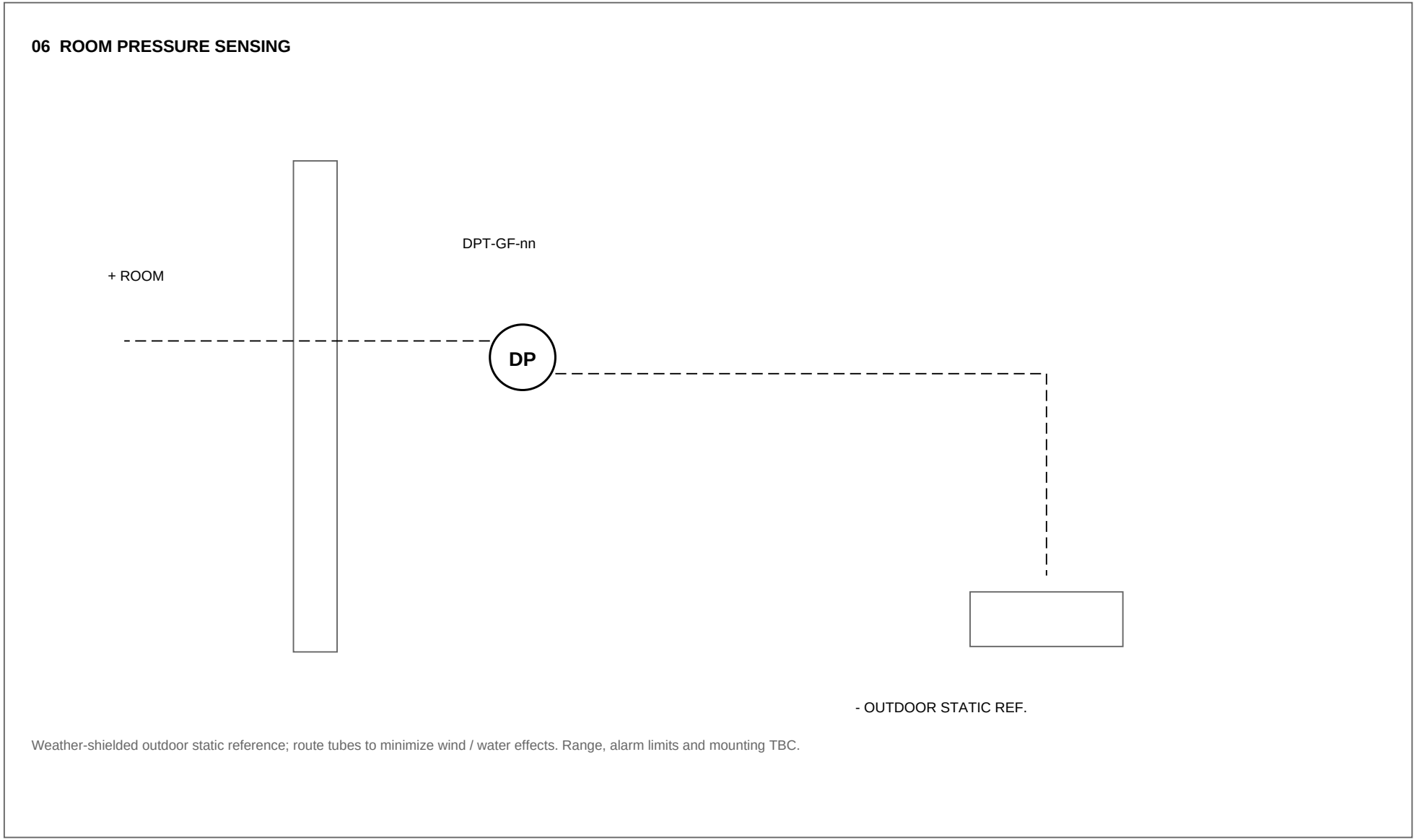
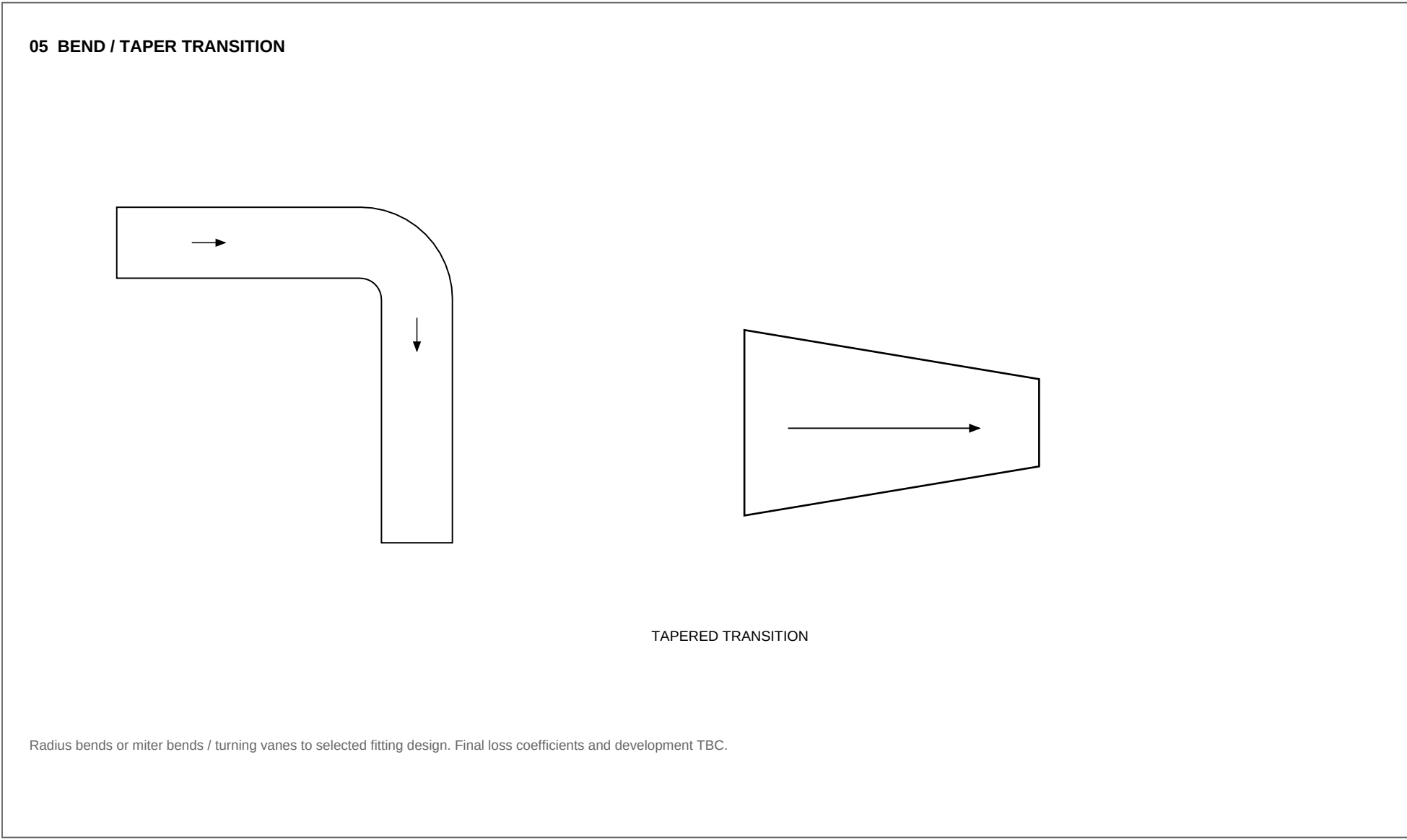
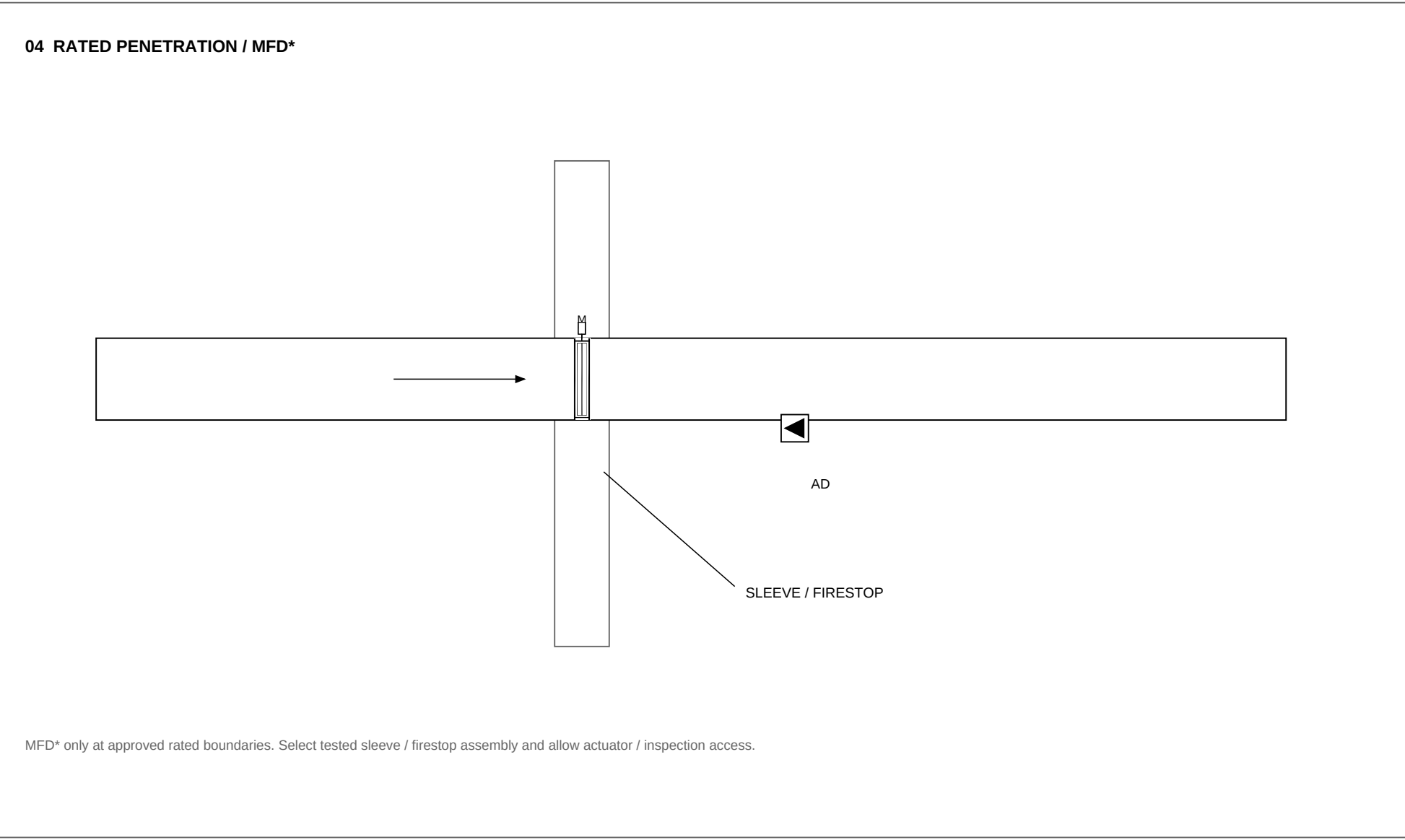
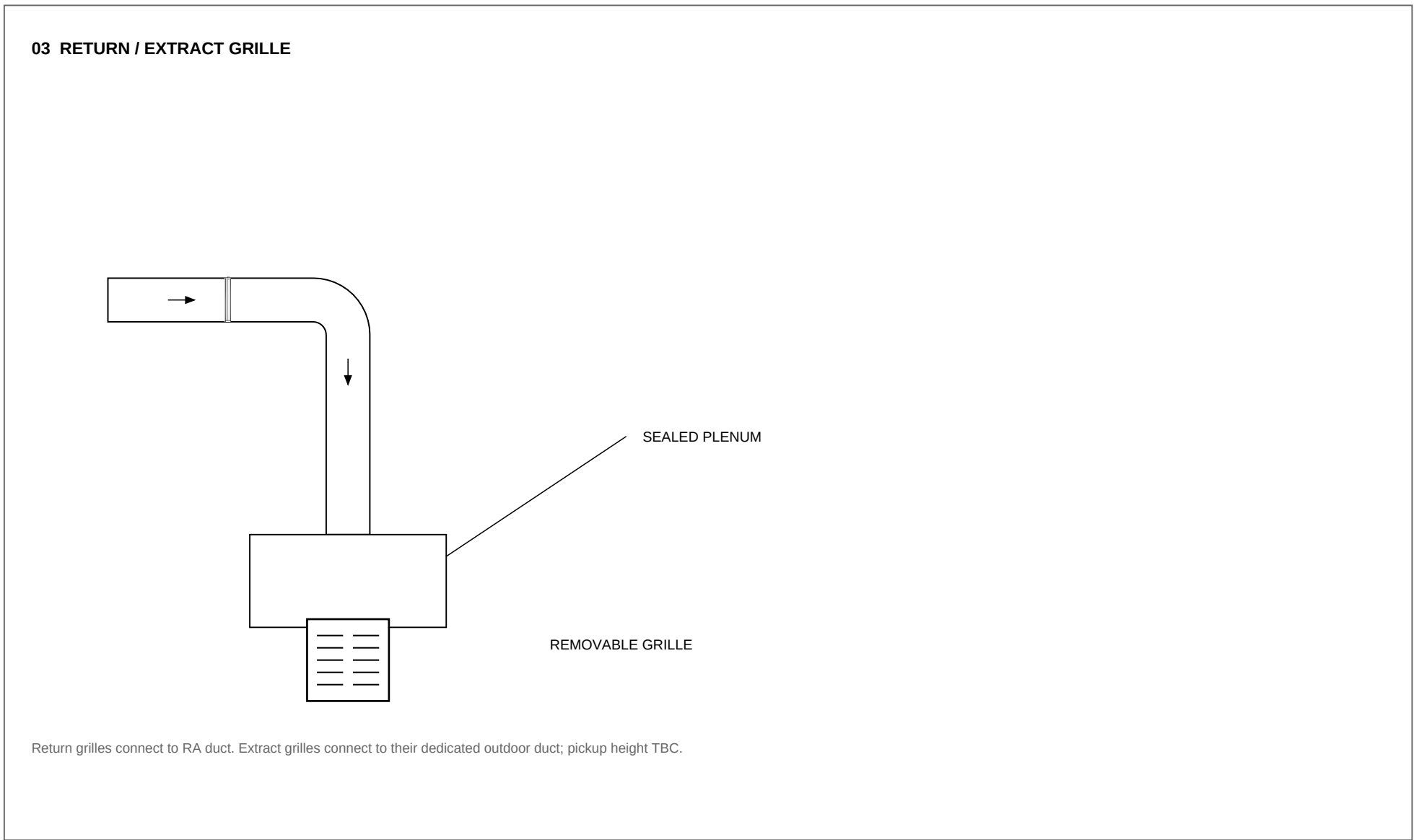
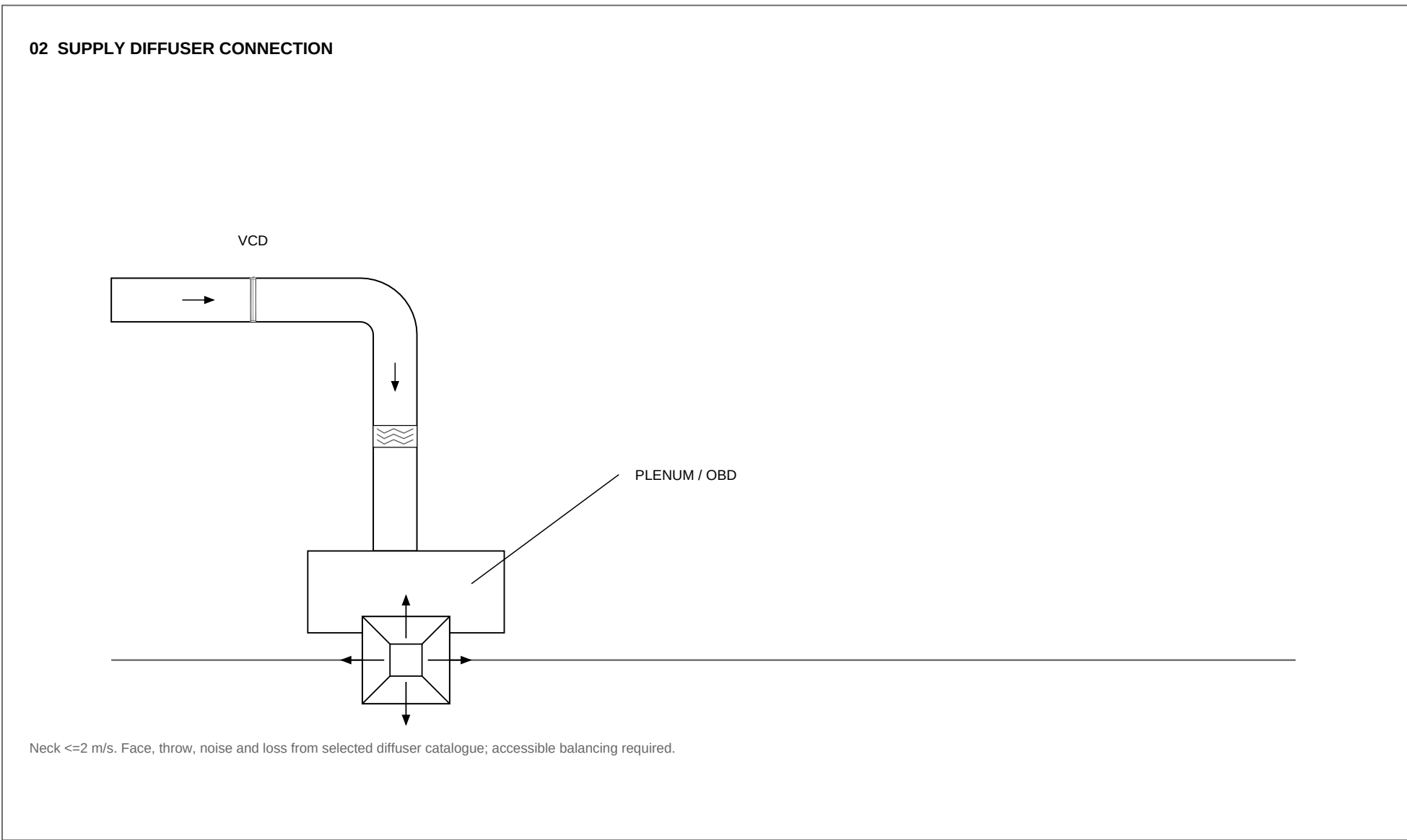
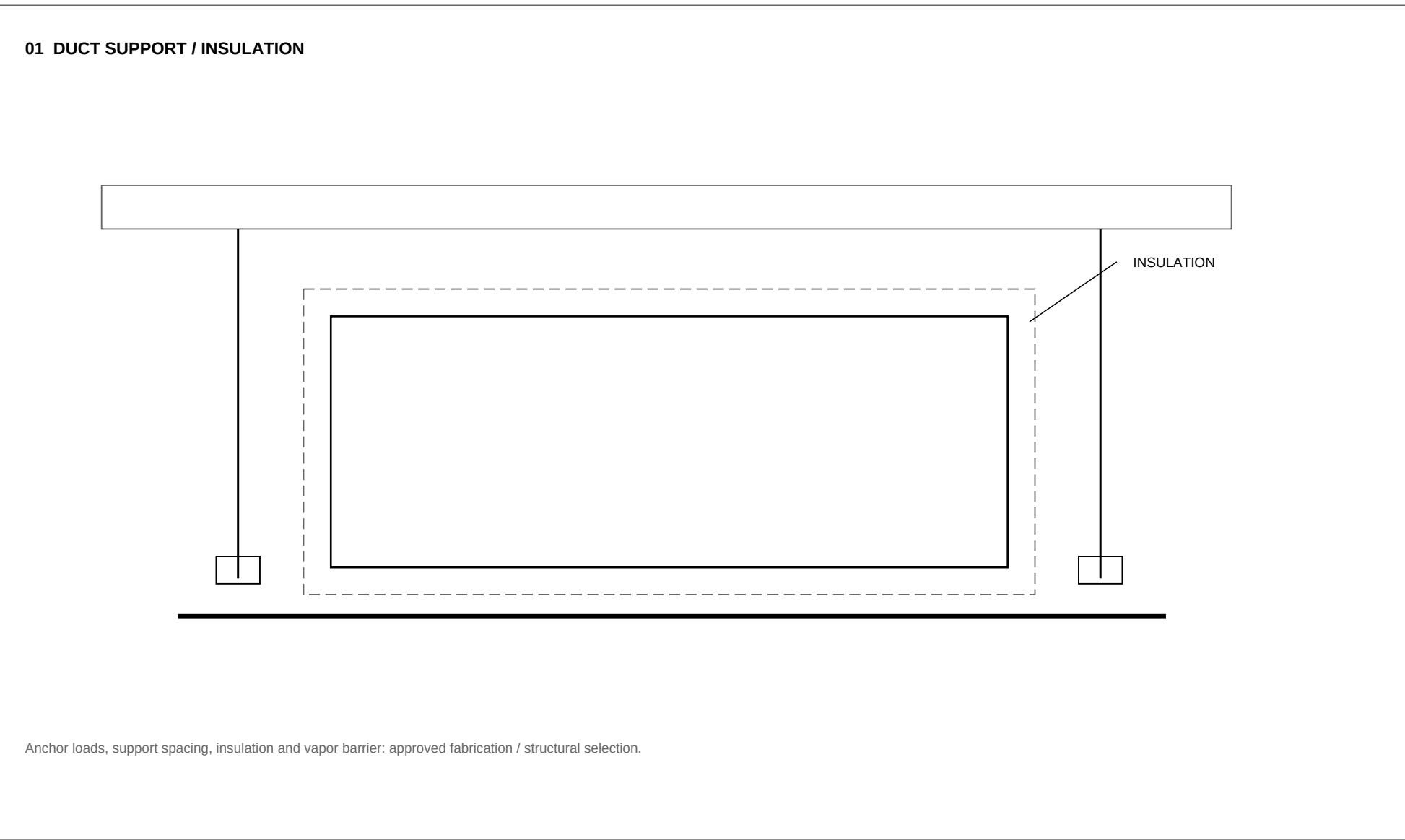
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Details are NTS engineering concepts. Duct colours / symbols match the plans; body dimensions are not manufacturer selections. No room pressure, fire rating or equipment clearance is certified by this drawing.

ADMINISTRATION BUILDING - DUCT AND TERMINAL SCHEDULES

MAIN AND ROOM BRANCH SECTIONS

Section	Size mm	L/s	m/s	Lh m*	Pa/m
SA-M01	1200x650	3697.80	4.74	2.40	0.252
SA-M02	1100x575	3033.00	4.80	4.08	0.294
SA-M03	650x750	2255.30	4.63	5.40	0.303
SA-M04	650x575	1806.10	4.83	1.55	0.386
SA-M05	650x300	921.50	4.73	6.80	0.599
SA-M06	500x250	503.30	4.03	0.90	0.575
SA-M07	450x250	457.10	4.06	2.10	0.611
SA-M08	400x200	288.90	3.61	2.30	0.619
SA-M09	300x200	141.30	2.35	1.60	0.325
RA-M01	1400x650	3006.68	3.30	3.50	0.122
RA-M02	1200x600	2401.22	3.34	4.10	0.140
RA-M03	600x800	1682.85	3.51	4.47	0.185
RA-M04	600x600	1292.99	3.59	3.20	0.228
RA-M05	350x575	467.73	2.32	3.30	0.152
RA-M06	350x300	108.86	1.04	1.70	0.052
SA-B02	500x350	664.80	3.80	3.00	0.399
RA-B02	600x400	605.46	2.52	2.50	0.157
SA-B04	350x400	449.20	3.21	4.22	0.330
RA-B04	500x350	389.86	2.23	0.93	0.151
SA-B08	600x400	884.60	3.69	6.67	0.313
RA-B08	500x650	825.26	2.54	3.83	0.130
SA-B05	600x350	777.70	3.70	12.88	0.349
RA-B05	500x525	718.36	2.74	16.00	0.168
SA-B06	450x300	418.20	3.10	1.23	0.324
RA-B06	350x550	358.86	1.86	0.62	0.104
SA-B09	200x150	46.20	1.54	1.88	0.231
SA-B07	300x250	168.20	2.24	7.28	0.254
RA-B07	350x250	108.86	1.24	6.03	0.081
SA-B03	300x200	147.60	2.46	2.58	0.351
SA-B01	300x200	141.30	2.35	8.57	0.325
EA-B09	250x200	64.89	1.30	6.70	0.123
EA-B03	300x200	147.60	2.46	1.90	0.351
EA-B01	TBC	TBC	TBC	TBC	TBC

* Lh is the source orthogonal horizontal route allowance. Final straight lengths, selected elbow development, risers / drops and fitting losses require coordination. Vertical lengths and BOD are unverified.

Sizes are clear internal dimensions. Catalogue free area, diffuser throw / noise, coil / filter / louver / silencer losses, fitting coefficients and OEM fan curves are pending. Confirmed fire stops the entire HVAC.

TERMINAL GROUPS

GF	Type	Qty	Each L/s	Neck mm	Face mm
02	SD	4	166.20	D350	600x600
02	RG	4	151.37	D300	500x500
05	SD	5	155.54	D325	600x600
05	RG	5	143.67	D300	500x500
04	SD	3	149.73	D325	600x600
04	RG	3	129.95	D300	500x500
08	SD	6	147.43	D325	600x600
08	RG	6	137.54	D300	500x500
06	SD	3	139.40	D300	600x600
06	RG	4	89.72	D300	500x500
07	SD	1	168.20	D350	600x600
07	RG	1	108.86	D300	500x500
09	SD	2	23.10	D150	300x300
09	EG	2	32.44	D150	300x300
03	SD	1	147.60	D325	600x600
03	EG	1	147.60	D300	500x500
01	SD	1	141.30	D300	600x600
01	EG	1	TBC	TBC	500x500

SYSTEM / DESIGN BASIS

Total supply: 3697.80 L/s. Assumed return: 3006.68 L/s.

Six return rooms: 59.34 L/s net outward allowance each, based on the accepted 0.01 m2 illustrative equivalent area.

Final room pressure, IAQ outdoor-air duty and revised HAP loads remain unverified. Current 420.70 L/s OA is below the 691.12 L/s assumed-balance makeup requirement.

All 122 duct sections, 77 fitting entries, 53 terminals, 176 components and 85 instruments are in the editable registers.

Rev10: routing / fitting topology revised, changed sizes and proposed terminal relocations are registered on D-021. Same-service overlaps and duplicated centerlines are rejected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DUCT AND TERMINAL SCHEDULES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-010

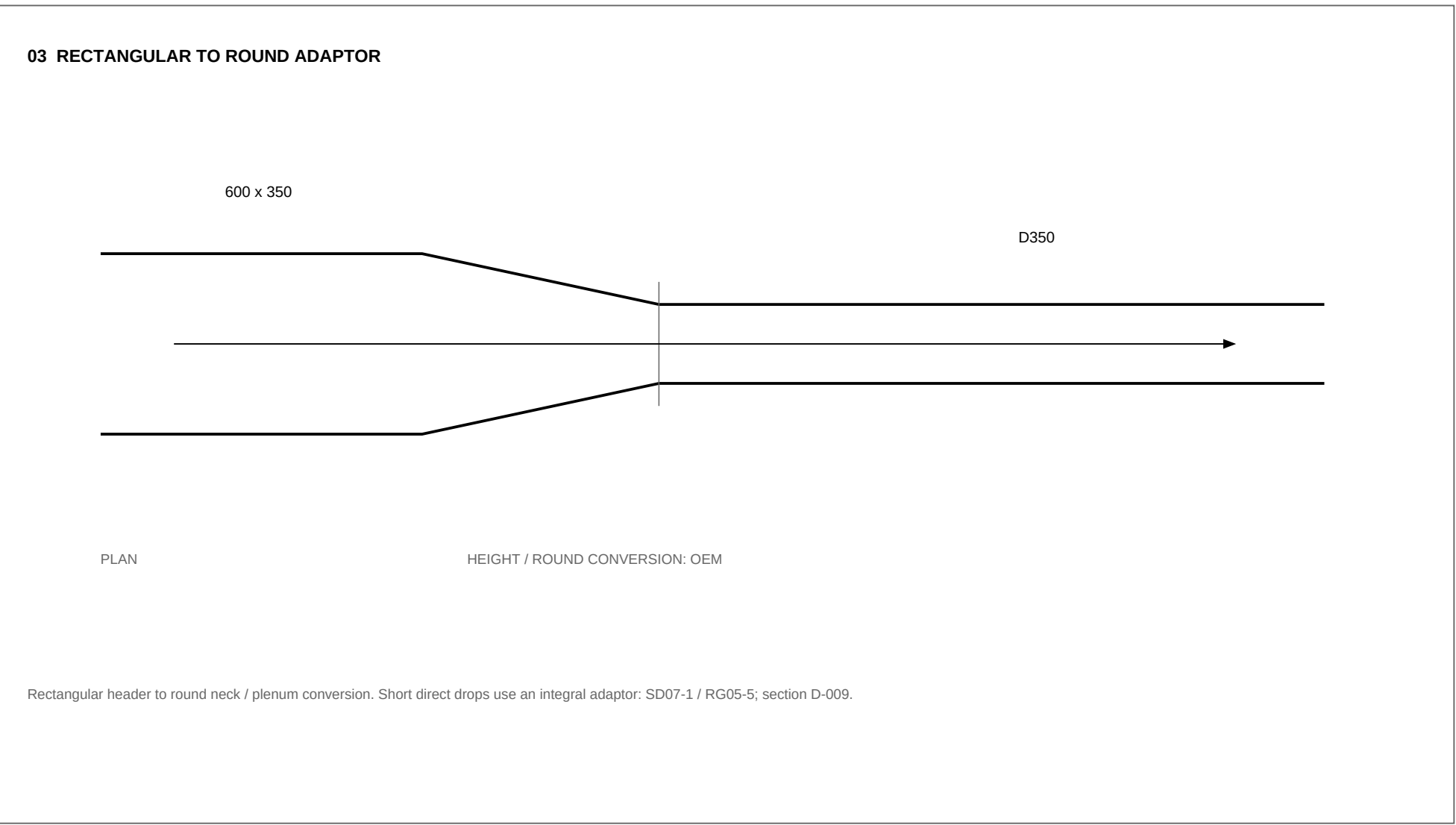
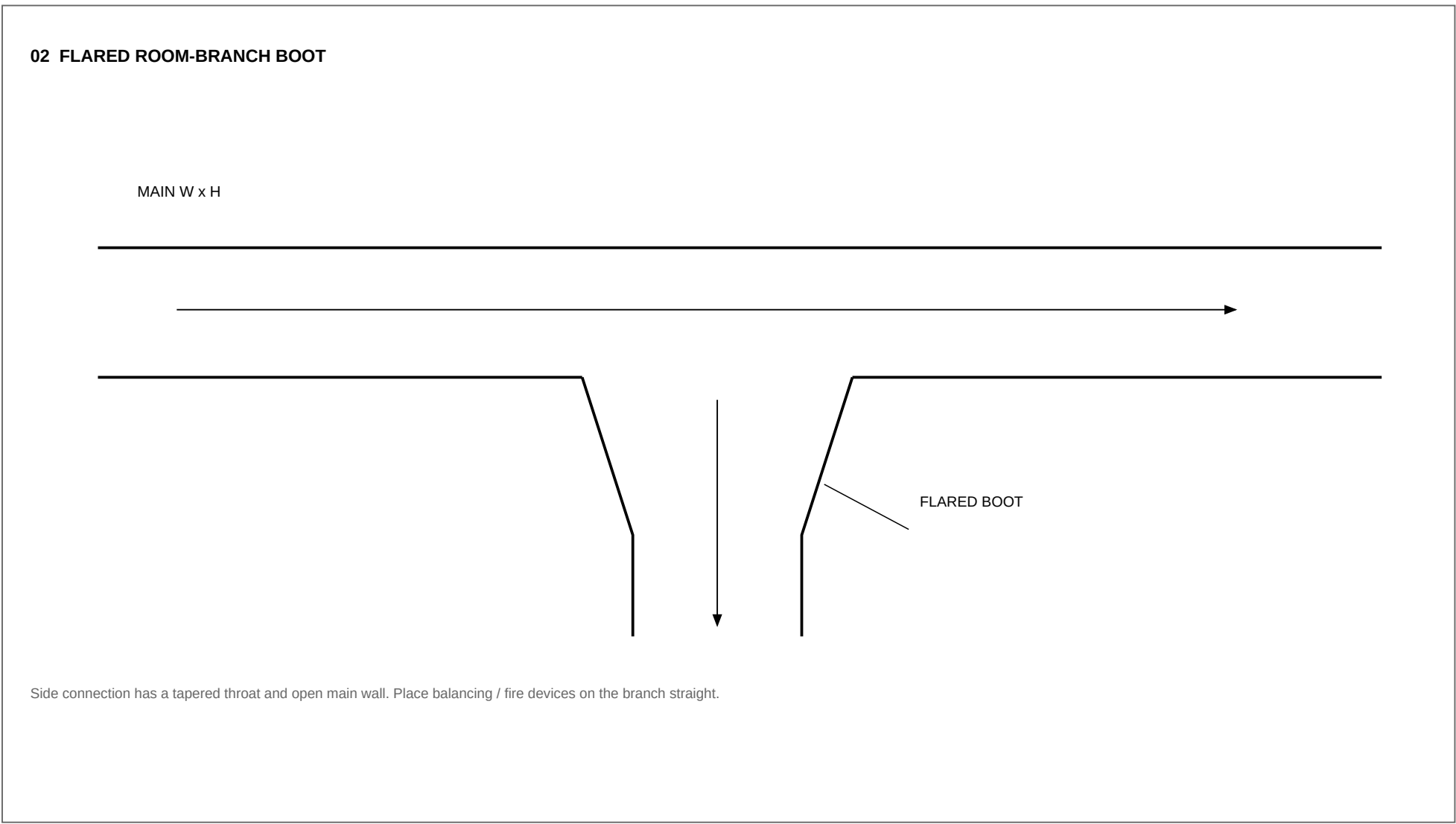
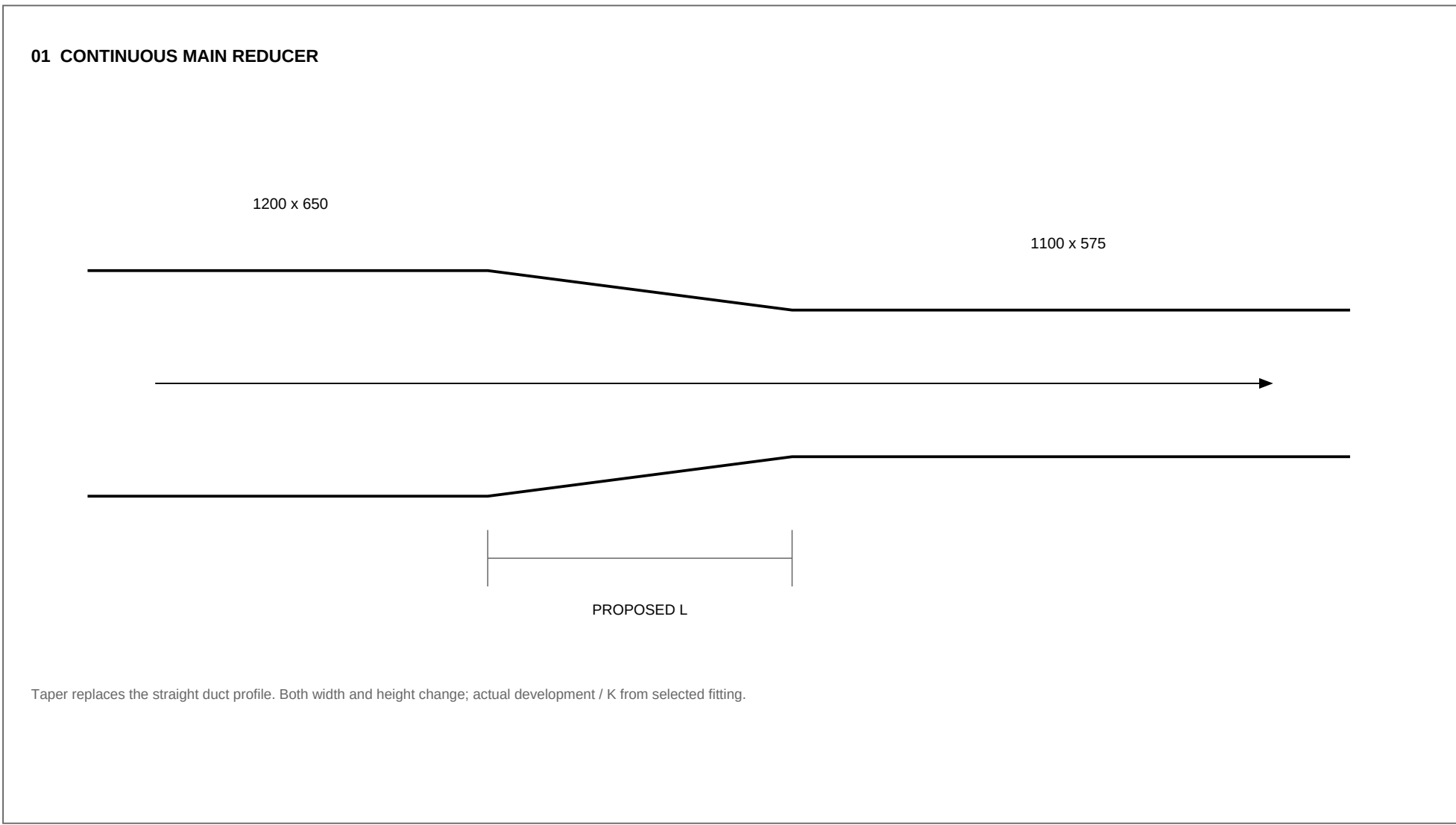
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REVISION

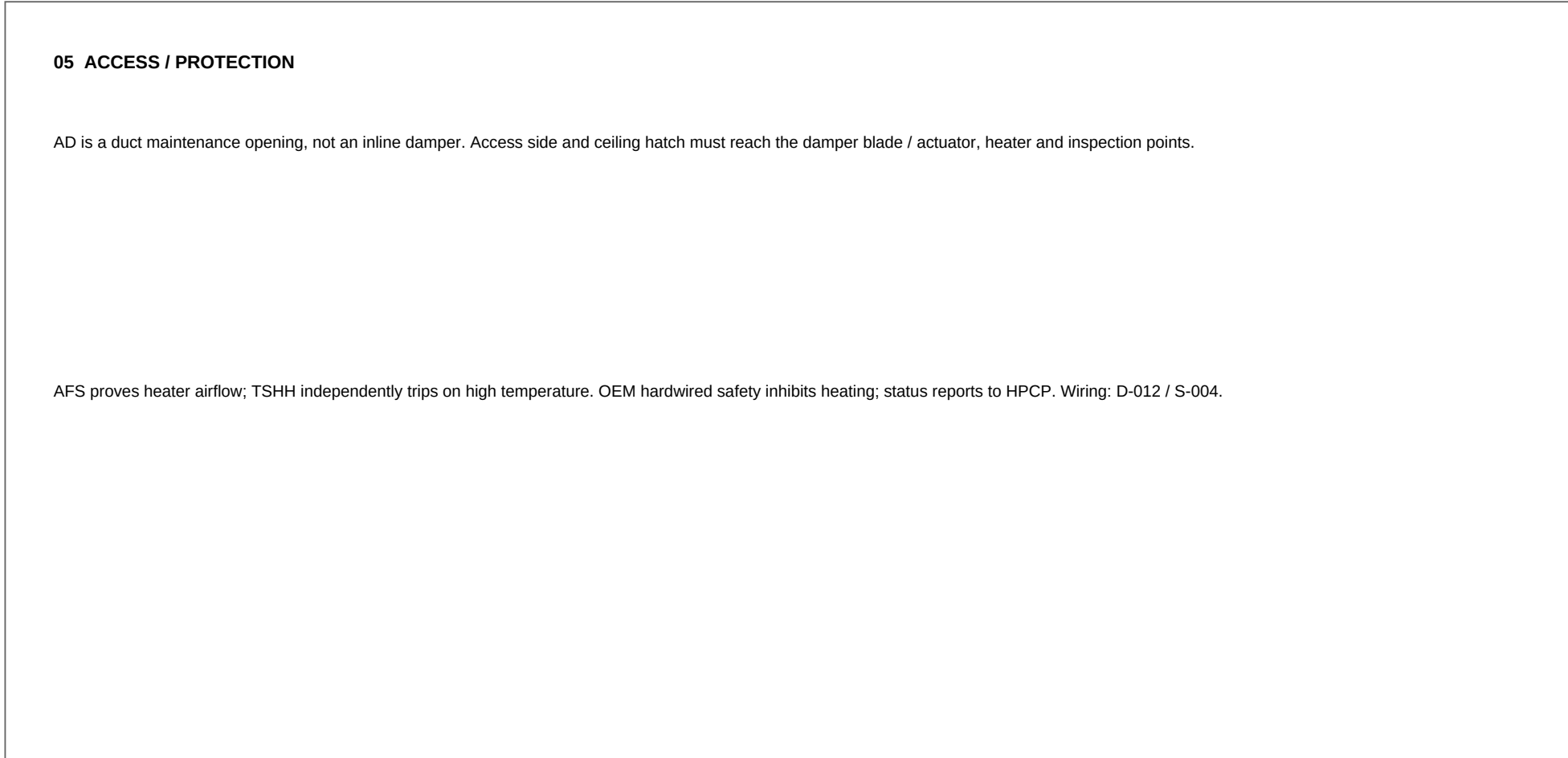
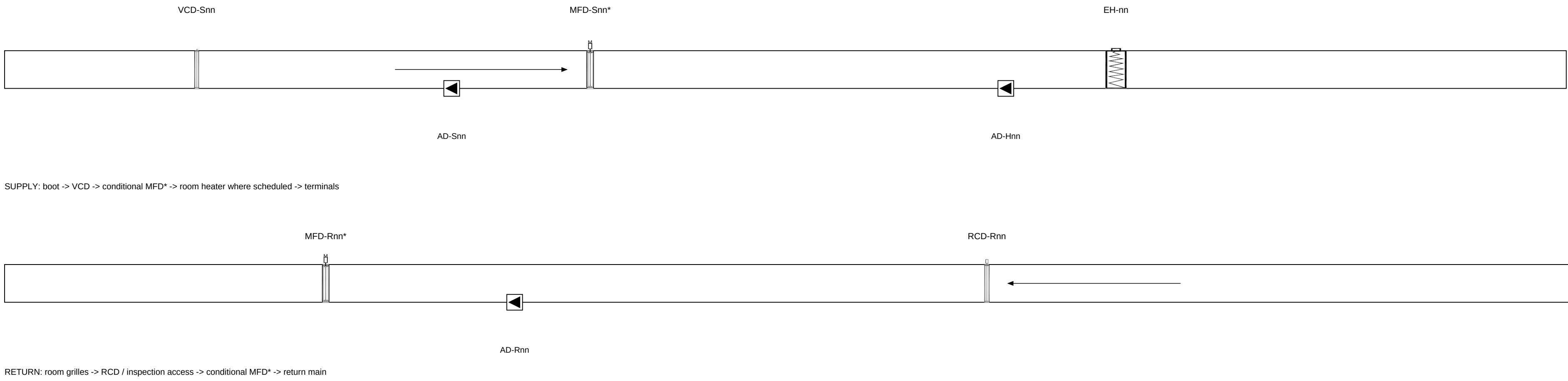
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04 TYPICAL BRANCH STATIONS - FLOW DIRECTION / ACCESS



MAIN TAPER SCHEDULE - PROPOSED GEOMETRY

Fitting	Inlet mm	Outlet mm	L mm*
SA-RED-01	1200x650	1100x575	350
SA-RED-02	1100x575	650x750	1125
SA-RED-03	650x750	650x575	350
SA-RED-04	650x575	650x300	350
SA-RED-05	650x300	500x250	375
SA-RED-06	500x250	450x250	350
SA-RED-07	450x250	400x200	350
SA-RED-08	400x200	300x200	350
RA-RED-01	1200x600	1400x650	500
RA-RED-02	600x800	1200x600	1500
RA-RED-03	600x600	600x800	350
RA-RED-04	350x575	600x600	625
RA-RED-05	350x300	350x575	350

ACCESS-DOOR TAGS / BRANCH MAP

GF	Supply access	Return access	Heater access
02	AD-S02	AD-R02	-
05	AD-S05	AD-R05	AD-H05
04	AD-S04	AD-R04	AD-H04
08	AD-S08	AD-R08	-
06	AD-S06	AD-R06	AD-H06
07	AD-S07	AD-R07	AD-H07
09	AD-S09	-	-
03	AD-S03	-	AD-H03
01	AD-S01	-	-
COM	AD-HCOM-01	-	Common heater

All 21 AD tags and device coordinates are registered. Door clear opening, sealing / insulation, access hatch and OEM working envelope remain to be selected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

TAPERED FITTINGS AND MAINTENANCE ACCESS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-011

SCALE: NTS AT A1

REVISION

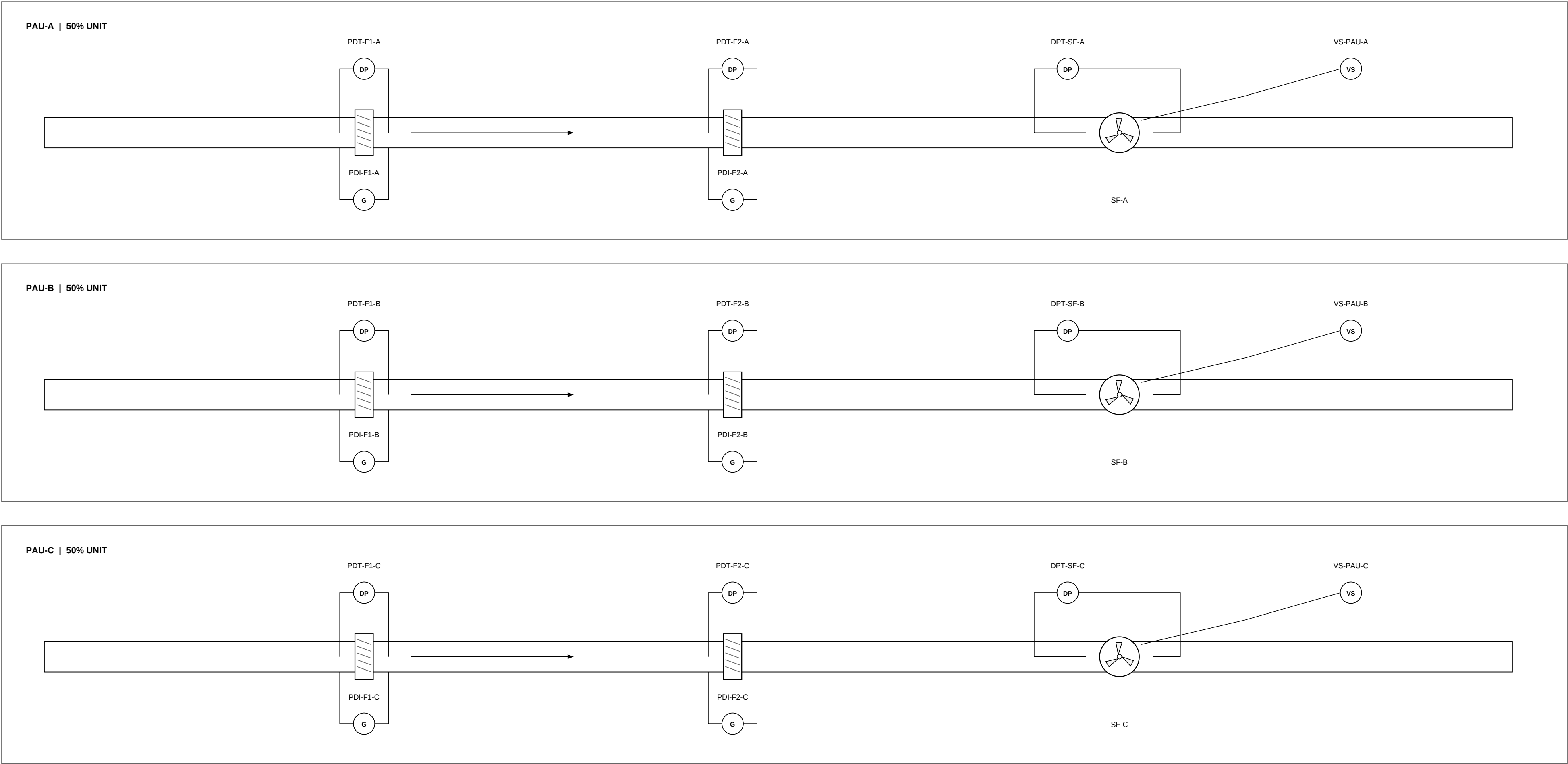
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07 OCT 2026

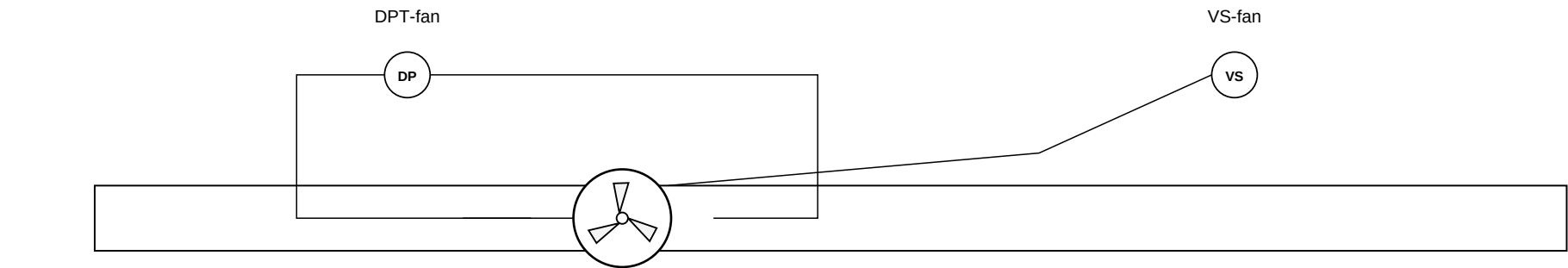
SHEET 11 OF 21

ADMINISTRATION BUILDING - PLANT INSTRUMENTATION AND HEATER PROTECTION

A EACH PAU: FILTER DIFFERENTIAL PRESSURE / FAN PROOF / VIBRATION



C EXTRACT / BASEMENT FAN PROOF

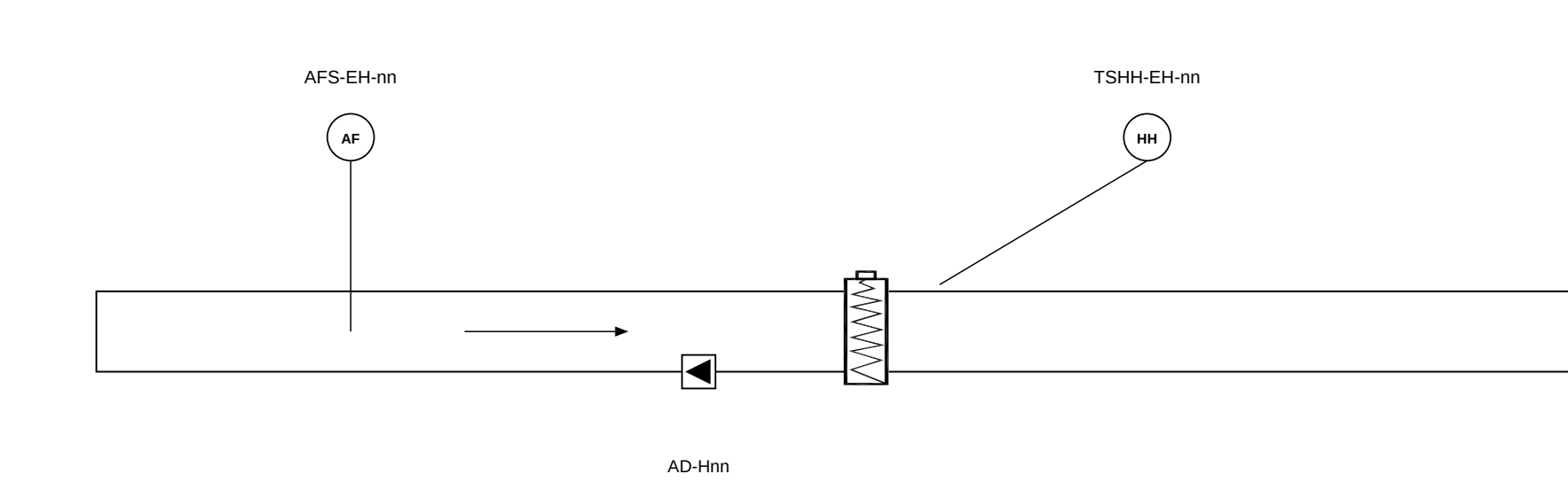


Tag suffixes: EF-03-01/02; EF-09-01/02; EF-01-01. Each has DPT- and VS- tags. Basement: SF-B1-01/02 and EF-B1-01/02 use the same convention.

Fan start / run / fault / ready / local-auto / E-stop and damper proofs are listed in I/O. CAG duty / quantity and basement duties remain pending.

Instrument glyphs are functional symbols, not equipment dimensions. AFS additions are explicit proposed OEM protection / monitoring provisions. Full tags: Excel / CSV registers. Room TT / RHT / DPT locations: D-007.

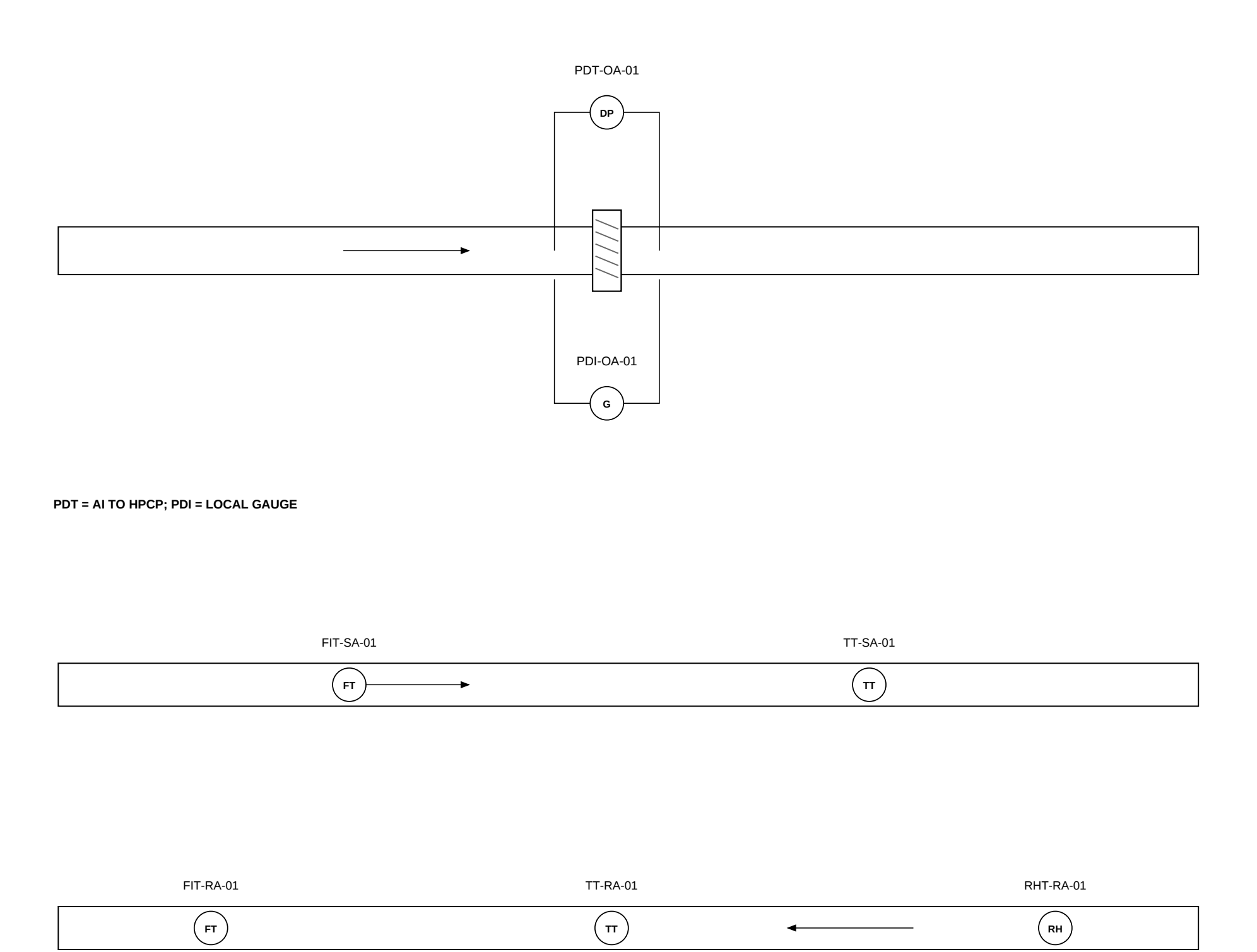
D HEATER AIRFLOW / HIGH-LIMIT SAFETY



nn = 04, 05, 06, 07, 03, COM-01. Six AFS DI points are retained from Rev07. Each heater retains independent TSHH; loss of proof / high limit switches heating off.

AFS proof technology, setpoint and OEM hardwired wiring to be selected. PLC demand cannot override local safety.

B OUTDOOR-AIR FILTER / COMMON DUCTS



Filter taps measure pressure across each filter, not between rooms. Fan DP proves the operating fan path; VS monitors fan / unit vibration. Alarm thresholds and OEM signal type TBC.

E PRELIMINARY HPCP I/O

Type	Points	Spare >=20%	Channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM: 3 interfaces; separate from wired channel totals. Signals report to dedicated HPCP-01 PLC / HMI-01 and BMS / SCADA. Final protocol and allocation TBC.

CONFIRMED FIRE: stop every HVAC fan and heater. No automatic fire extraction / post-fire purge. Reset and damper fire positions follow the approved cause-and-effect.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PLANT INSTRUMENTATION AND HEATER PROTECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-012

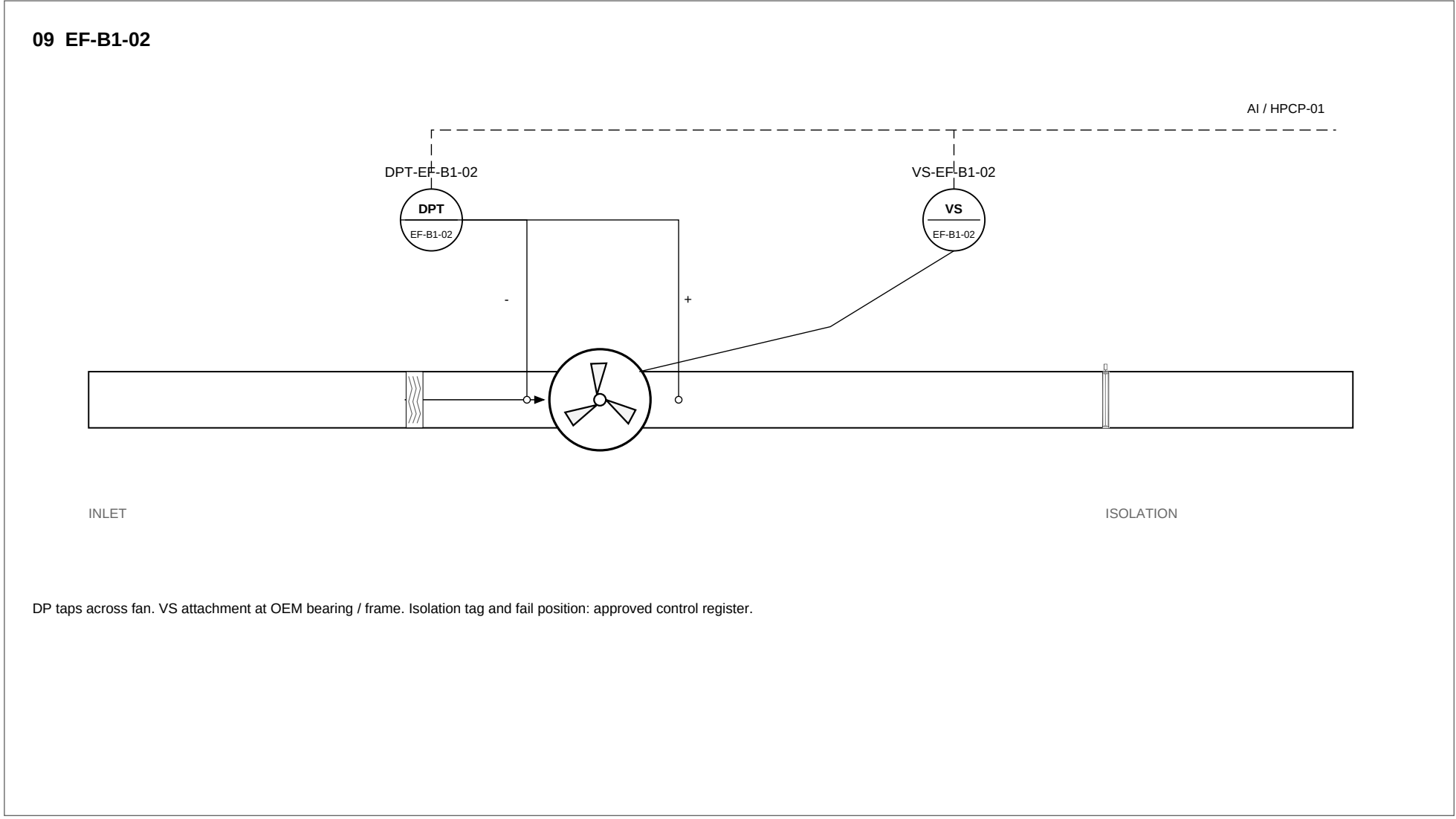
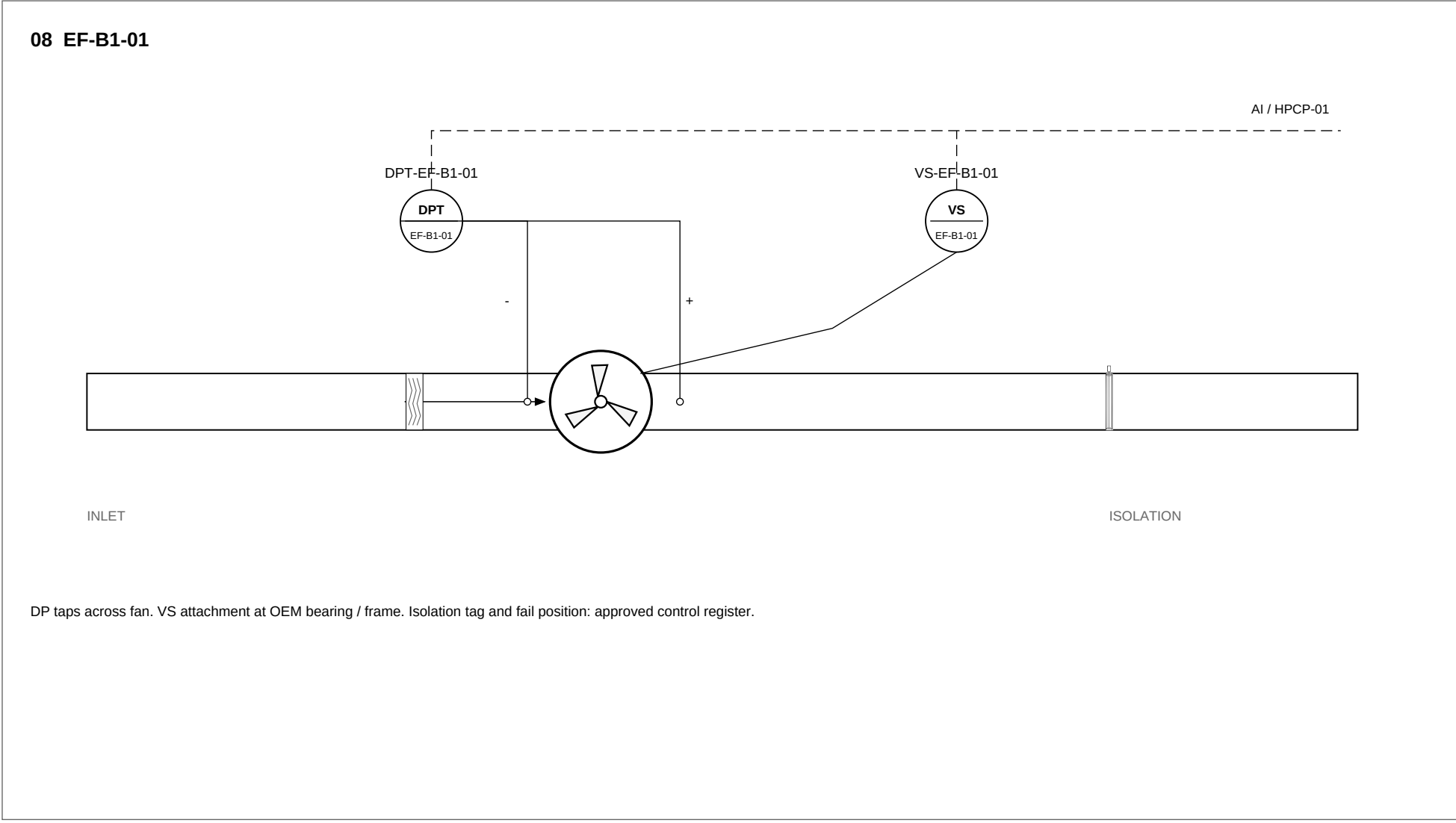
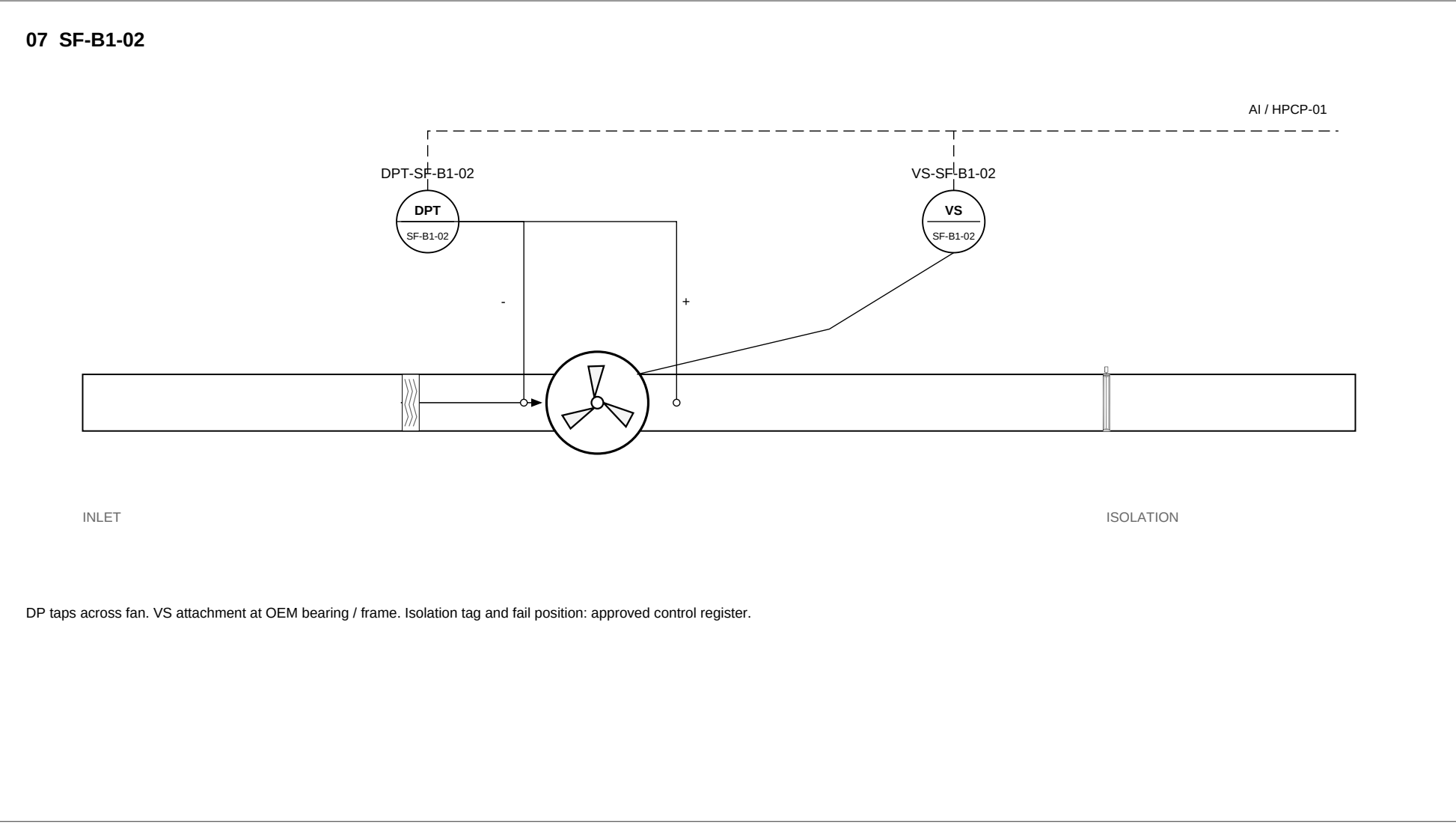
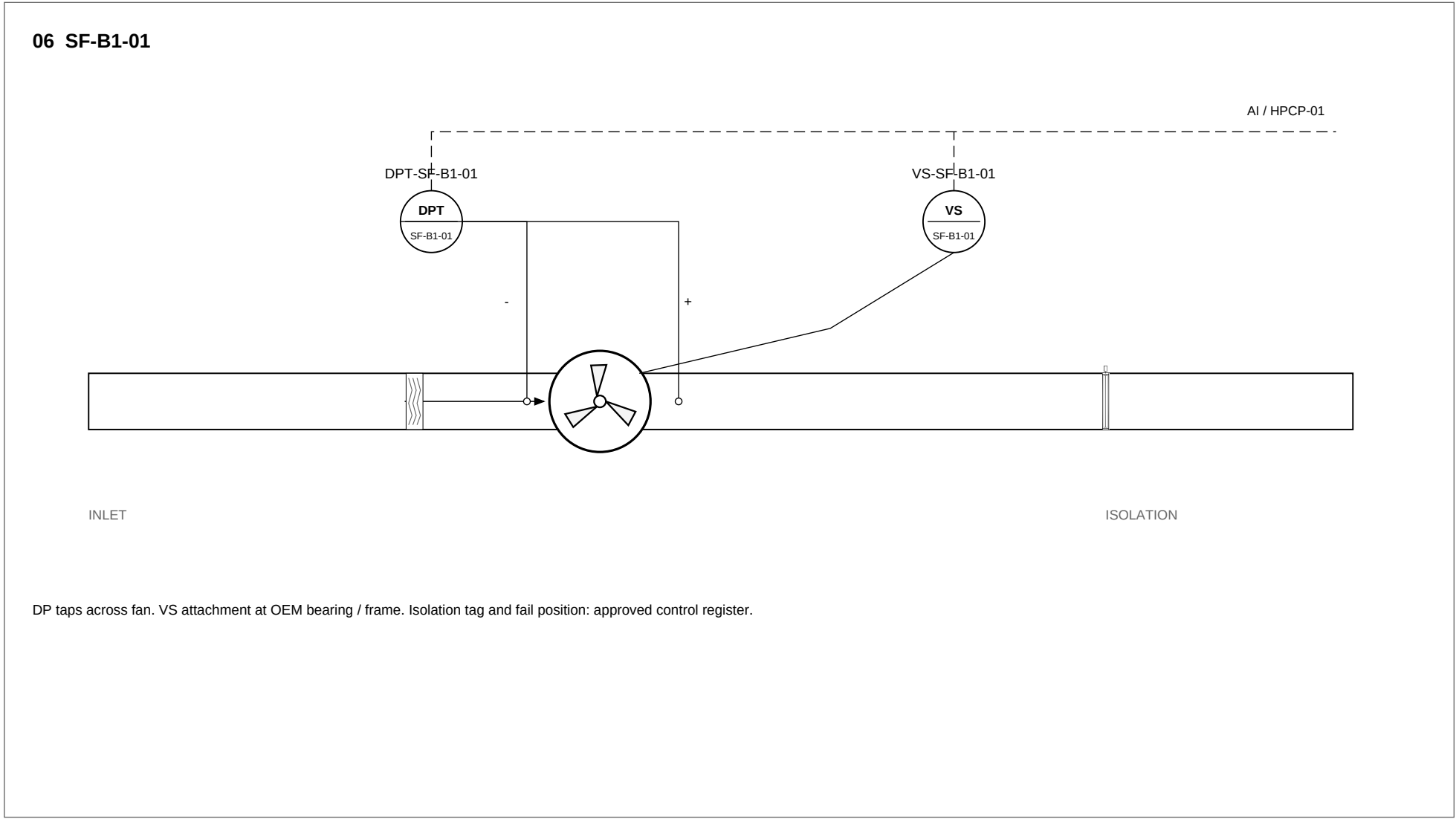
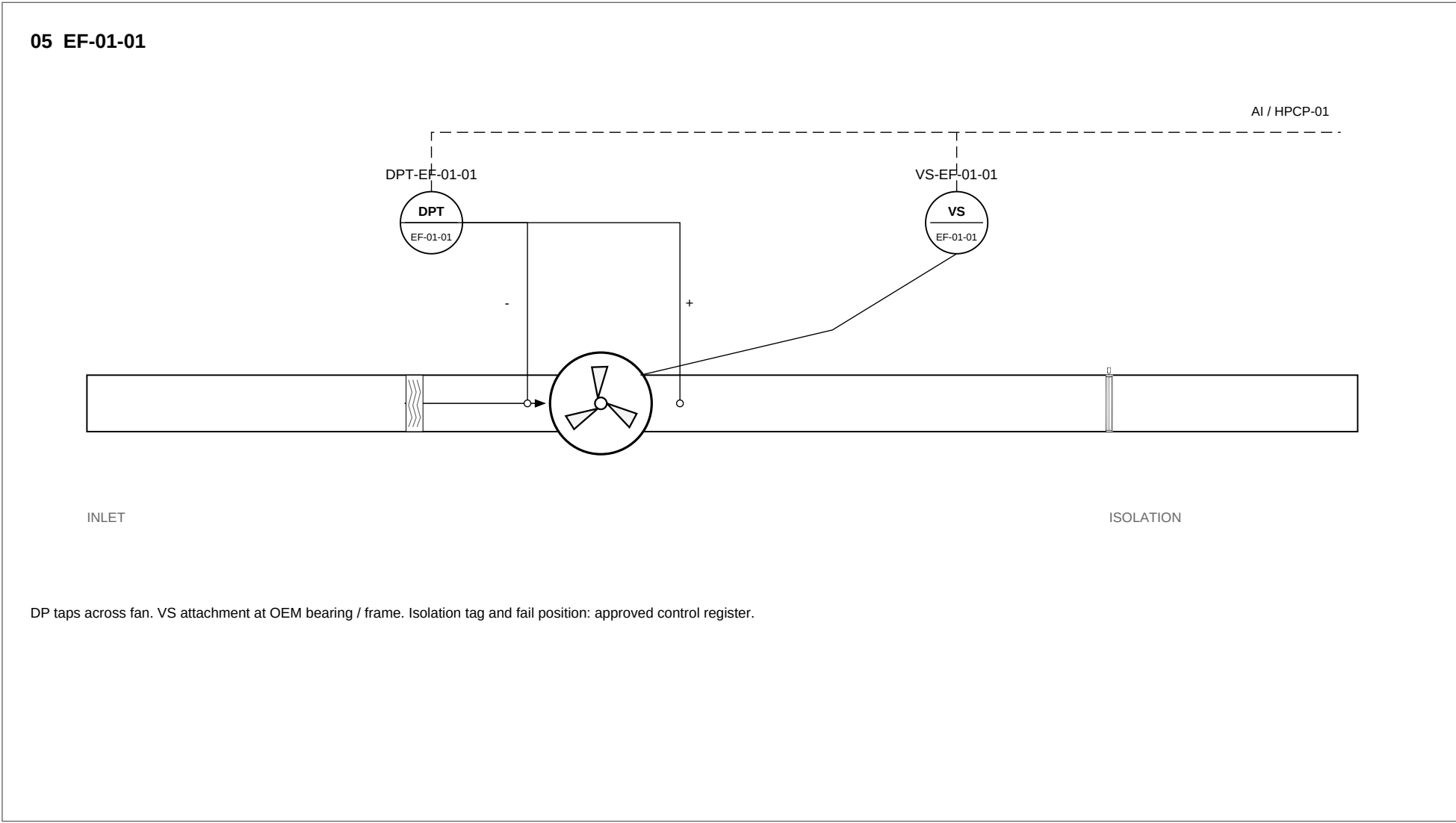
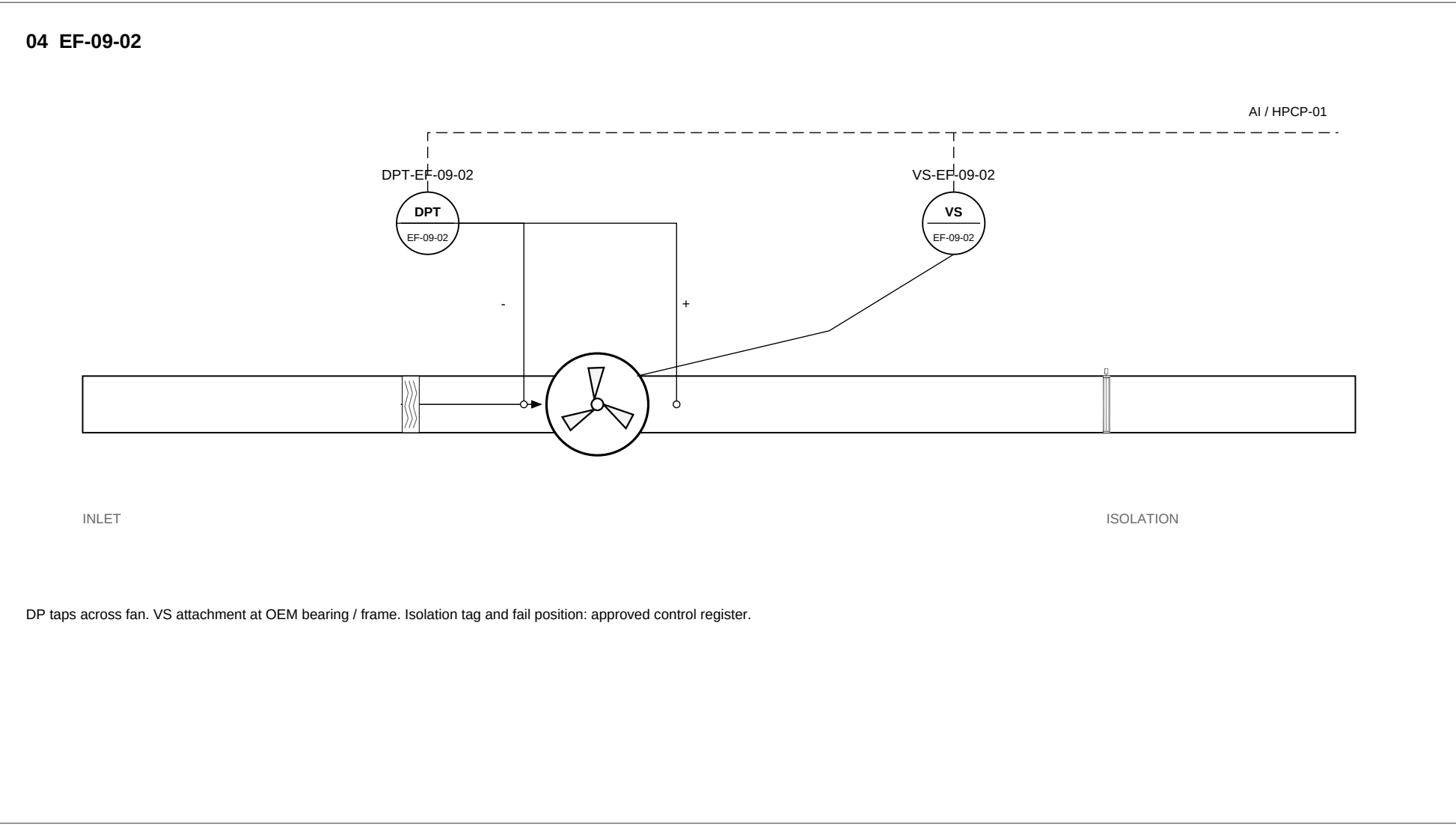
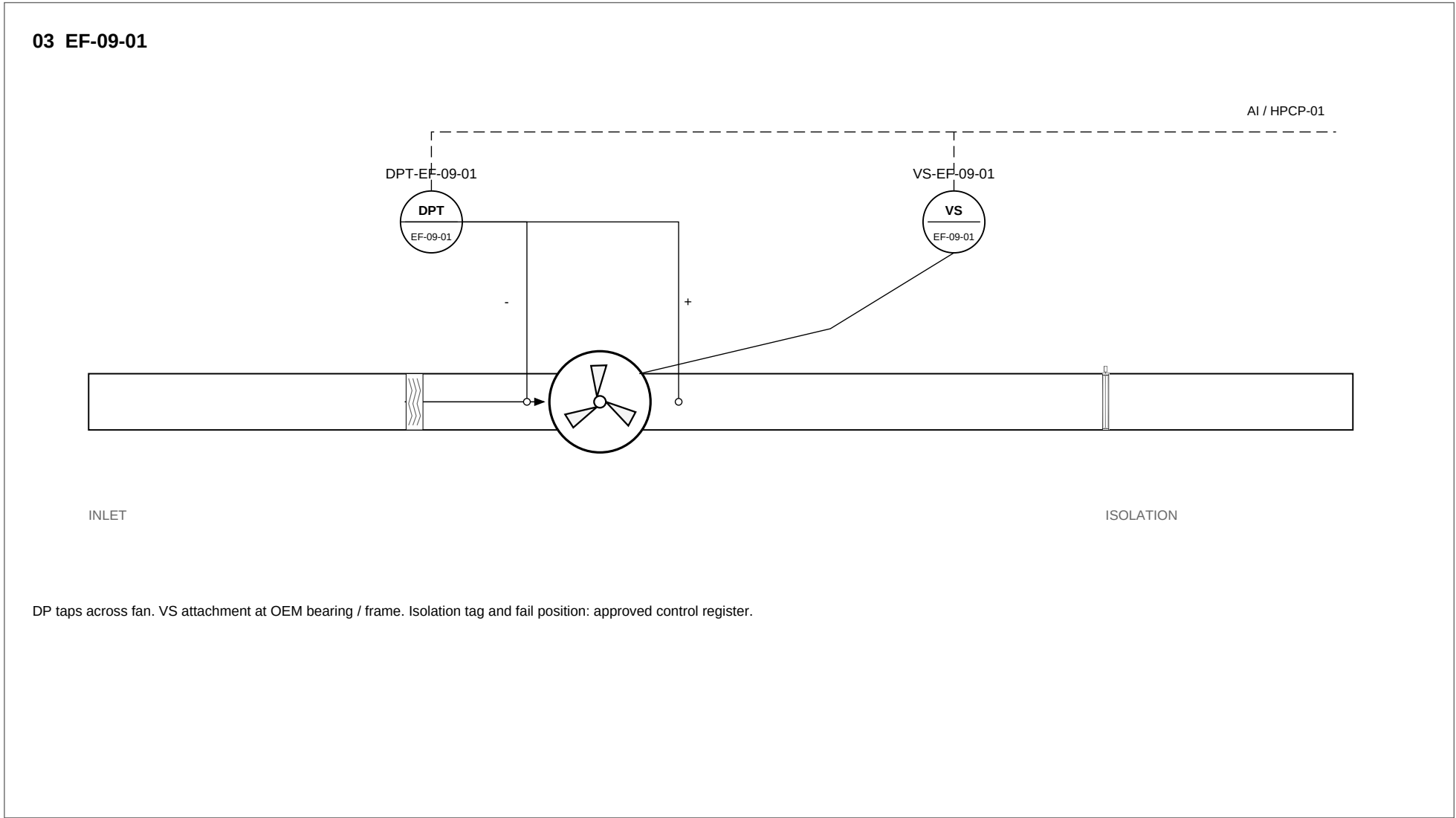
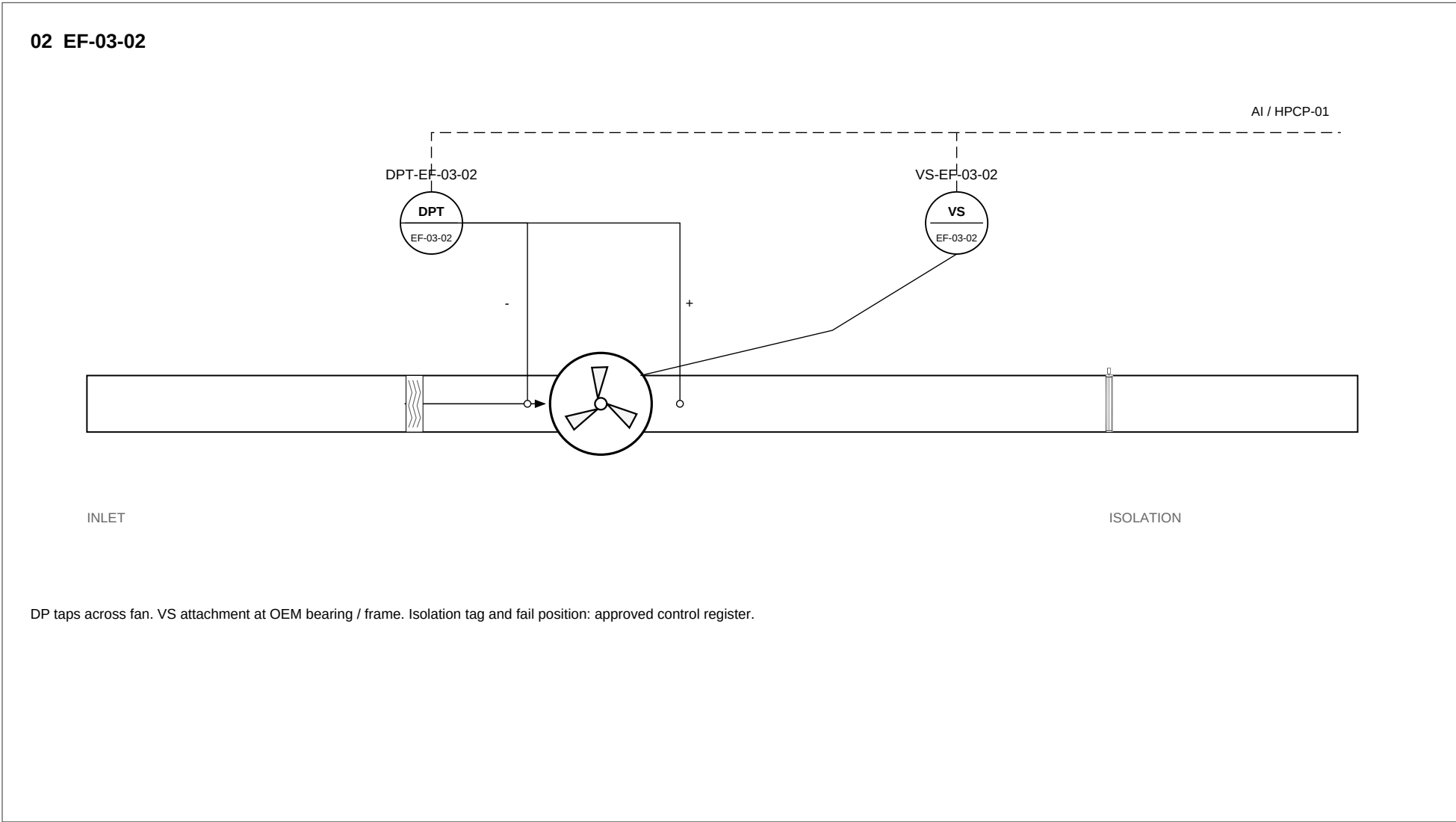
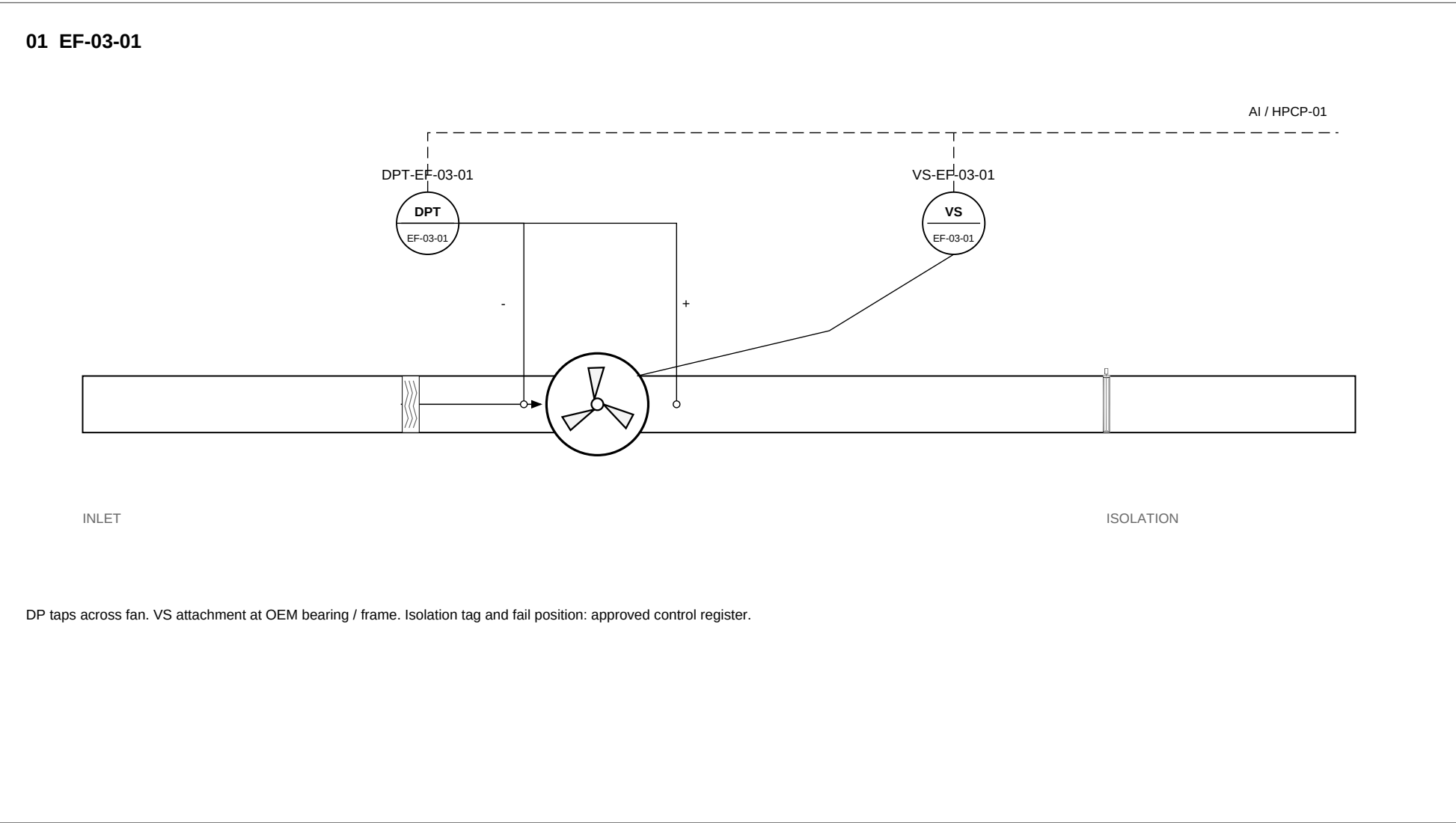
SCALE: NTS AT A1

REVISION

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SHEET 12 OF 21



NTS functional arrangements. Basement equipment positions are not shown on the Administration plan. CAG fan duty / quantity, tap locations, alarm limits and vibration mountings require OEM / I&C confirmation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

EXTRACT AND BASEMENT FAN INSTRUMENTATION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-013

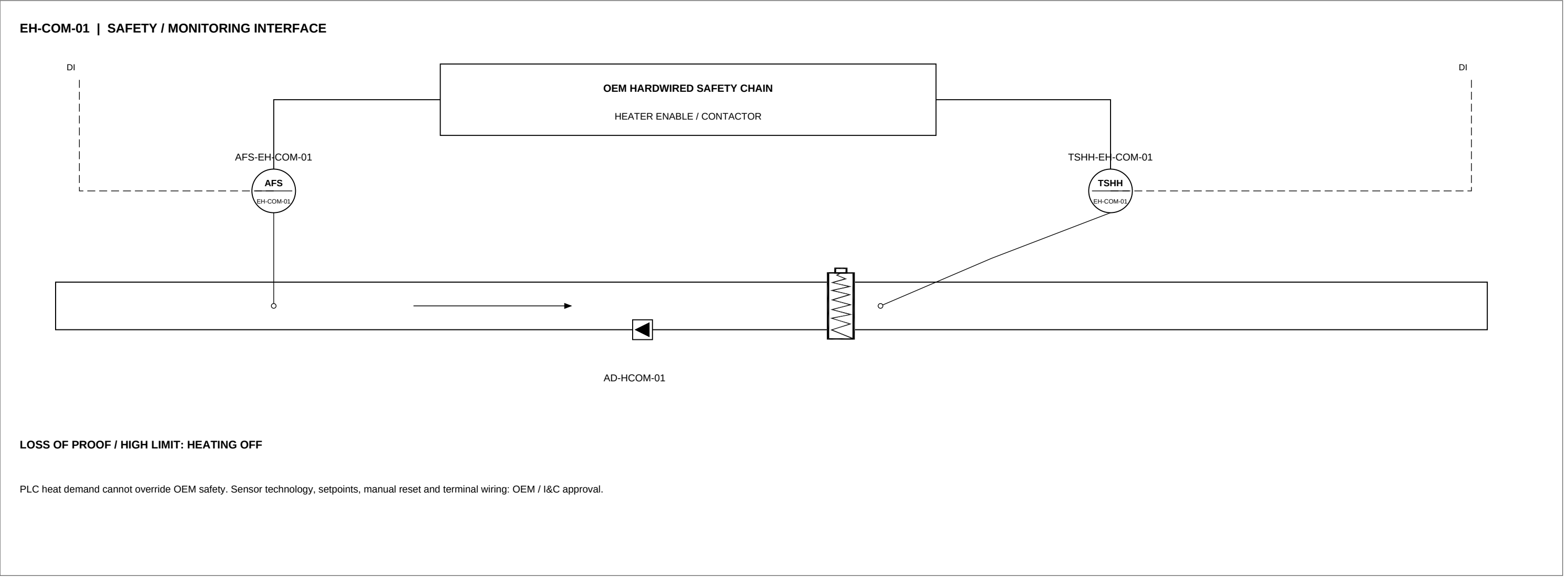
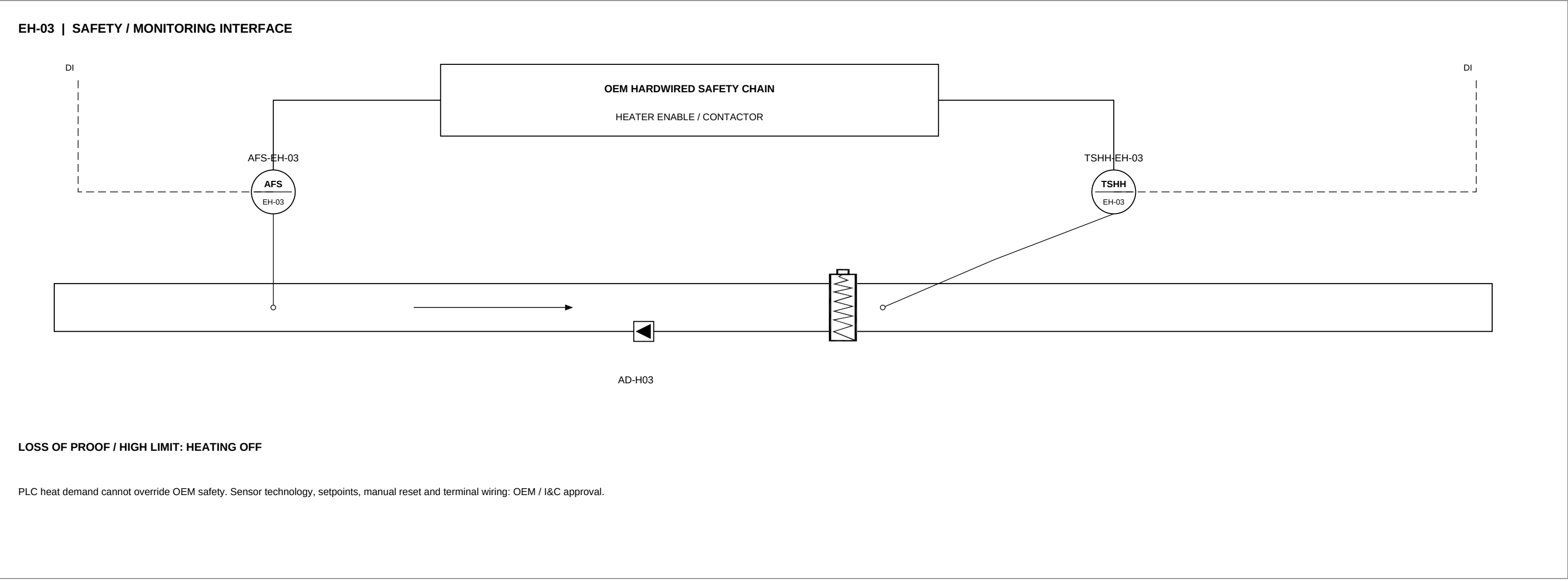
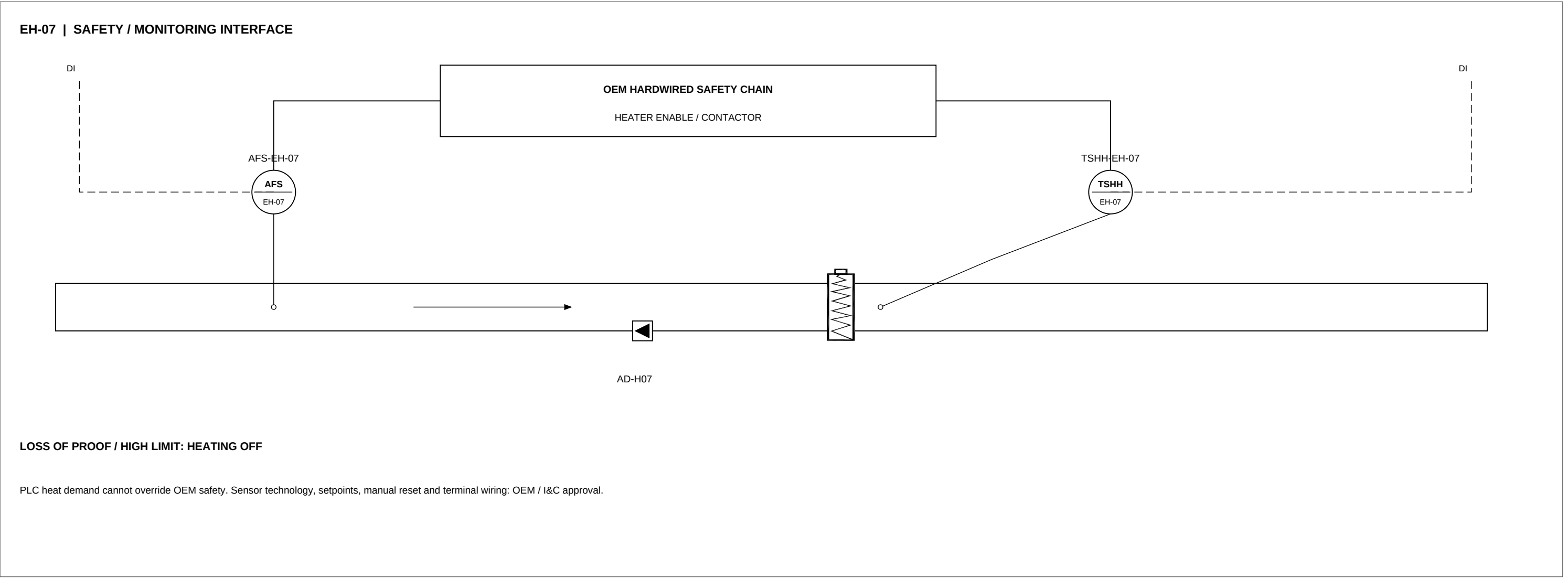
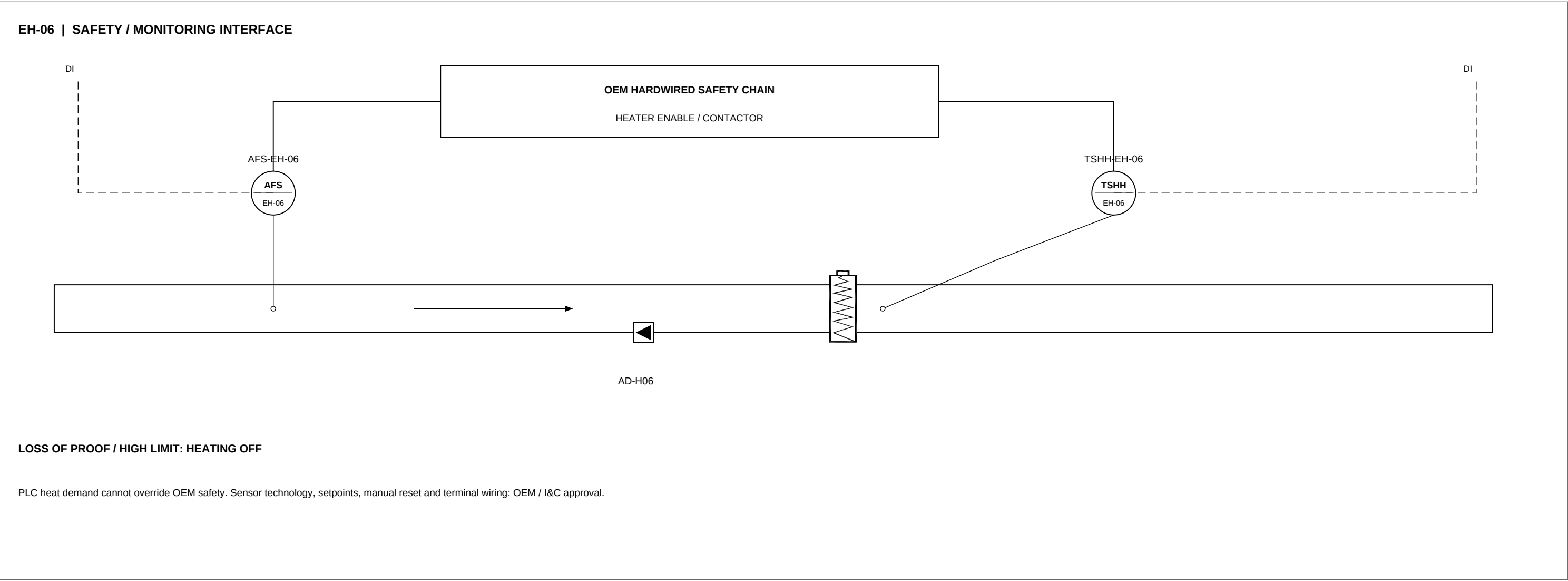
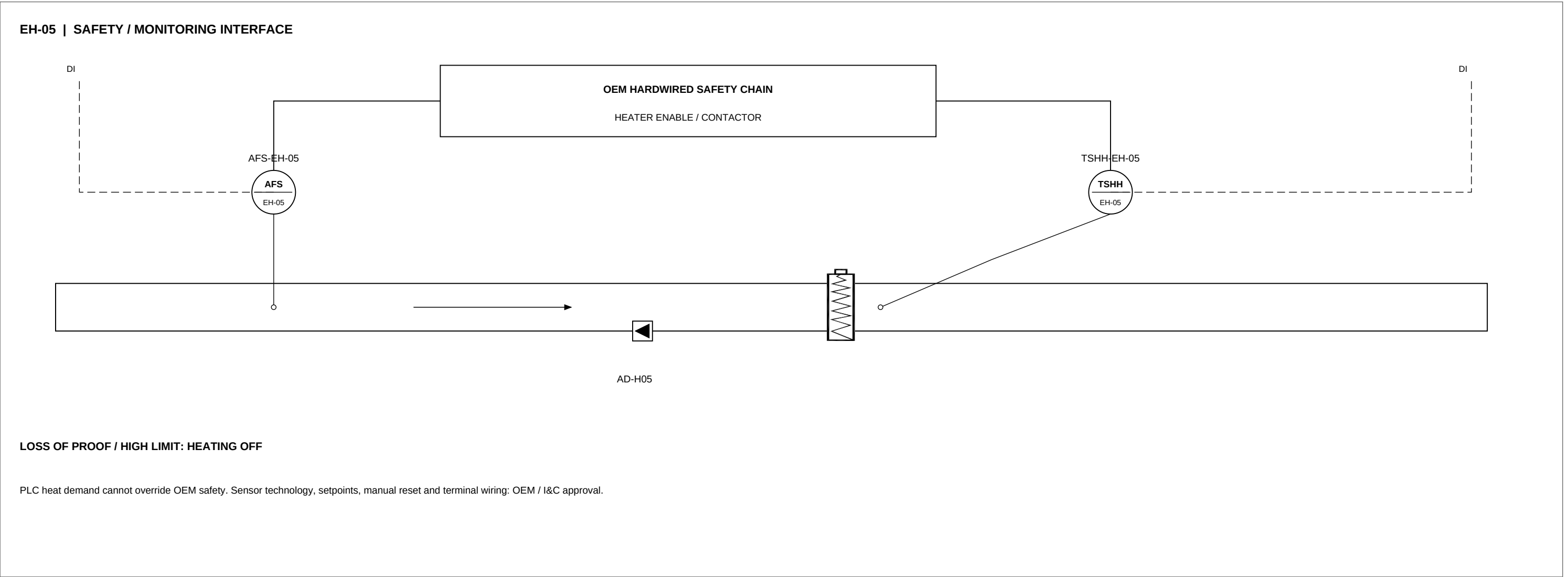
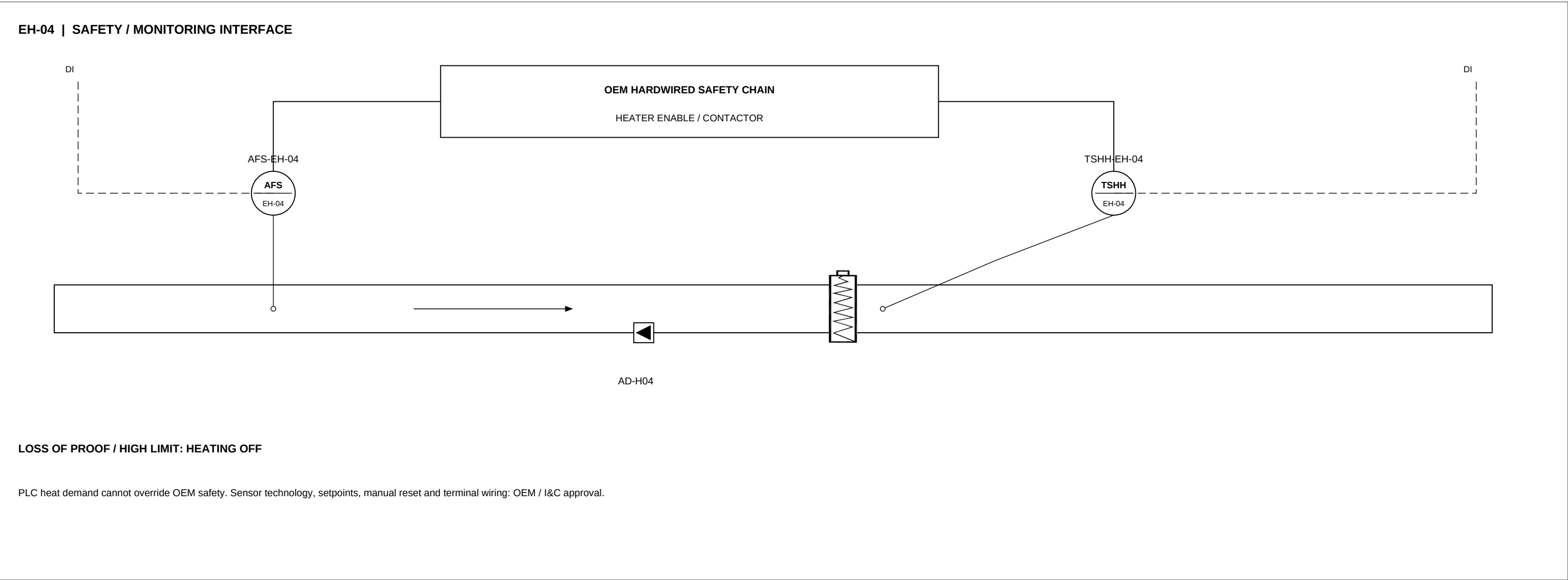
SCALE: NTS AT A1

REVISION

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SHEET 13 OF 21



Functional interfaces, not certified wiring diagrams. Six airflow-proof DI and six high-limit DI tags are retained. Protection circuits must be independently implemented and approved by the heater manufacturer.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ELECTRIC HEATER PROTECTION AND CONTROL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-014

SCALE: NTS AT A1

REVISION

10

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SHEET 14 OF 21

01 INSTRUMENT / SIGNAL CONVENTIONS

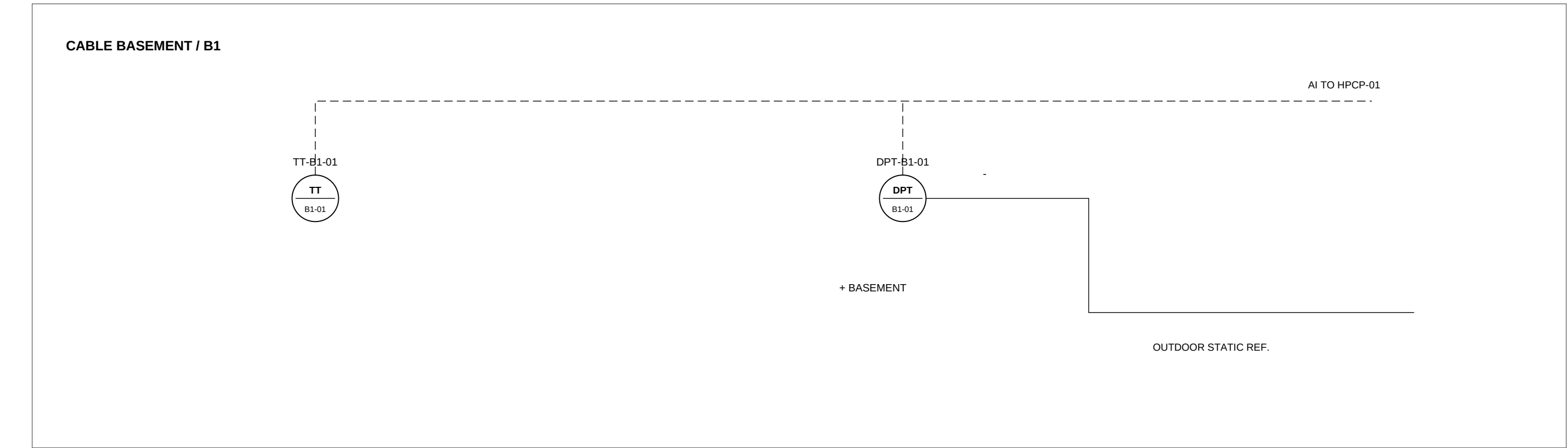
TT	Temperature transmitter
RHT	Humidity transmitter
DPT	Differential pressure transmitter
FIT	Airflow transmitter
POT	Remote filter DP transmitter
POI	Local differential pressure indication
VS	Vibration sensing
AFS	Airflow proving switch
TSHH	Heater high-temperature limit

Sensing / physical interface (route schematic)

Electrical / communications signal (schematic)

AI / AO / DI / DO are point classes, not selected signal ranges. 4-20 mA, contacts, network protocol, cable types and terminal numbers are not inferred.

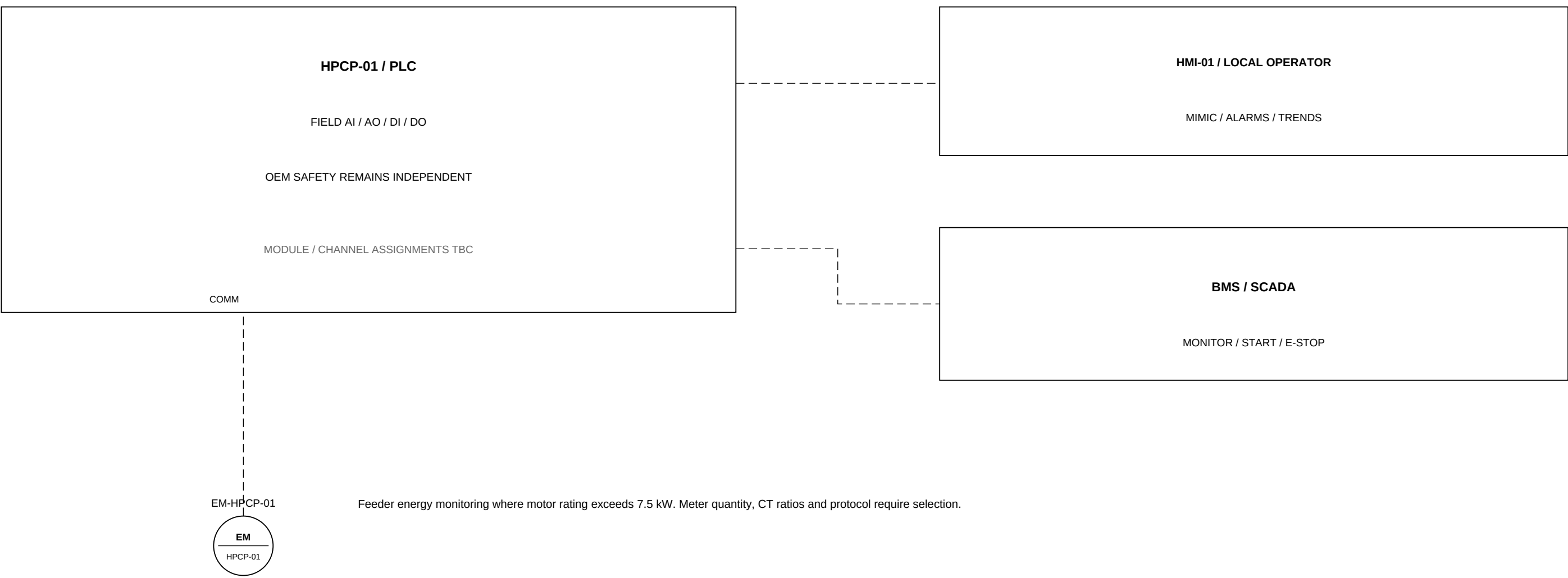
02 BASEMENT FIELD SENSING - PROPOSED



Basement temperature, volume, fan duties and actual pressure target remain open. Fan instrumentation is fully tagged on D-013; no basement architectural layout was supplied.

All controls shown are functional design intent. Approved OEM wiring and project I&C schedules govern implementation. D-016 provides an instrument-to-drawing cross-reference for every one of the 85 scheduled instruments.

03 HPCP / HMI / BMS INTERFACE



04 POINT CLASSES / MINIMUM SPARE

Class	Scheduled	20% spare	Min. channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM interfaces are separate from hardwired channels. Local PDI instruments have no PLC channel. Additional proposals are identified in the instrument index, D-016.

05 FIRE / STARTUP DESIGN INTENT

Confirmed fire: stop every HVAC fan and heater. Restart after fire ESD requires local manual reset. Damper fire positions are governed by the approved cause-and-effect, not this drawing.

Normal operation: two duty 50% PAUs with an available alternate. Prove selected air paths before starting fans. Duty rotation, changeover, timing and final permissives require approved OEM logic.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-015

SCALE: NTS AT A1

REVISION

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ADMINISTRATION BUILDING - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

Instrument tag	I/O	Service / location	Status	D-sheet
TT-GF-02	AI	GF-02 Elec. panel	Required	007
RHT-GF-02	AI	GF-02 Elec. panel	Required	007
DPT-GF-02	AI	GF-02 Elec. panel	Proposal	007
TT-GF-04	AI	GF-04 Meeting	Required	007
RHT-GF-04	AI	GF-04 Meeting	Required	007
DPT-GF-04	AI	GF-04 Meeting	Proposal	007
TSHH-EH-04	DI	EH-04	OEM safety	014
TT-GF-08	AI	GF-08 Telecom	Required	007
RHT-GF-08	AI	GF-08 Telecom	Required	007
DPT-GF-08	AI	GF-08 Telecom	Proposal	007
TT-GF-05	AI	GF-05 Operator	Required	007
RHT-GF-05	AI	GF-05 Operator	Required	007
DPT-GF-05	AI	GF-05 Operator	Proposal	007
TSHH-EH-05	DI	EH-05	OEM safety	014
TT-GF-06	AI	GF-06 Corridor	Required	007
RHT-GF-06	AI	GF-06 Corridor	Required	007
DPT-GF-06	AI	GF-06 Corridor	Required	007
TSHH-EH-06	DI	EH-06	OEM safety	014
TT-GF-09	AI	GF-09 Toilets	Required	007
RHT-GF-09	AI	GF-09 Toilets	Required	007
DPT-GF-09	AI	GF-09 Toilets	Proposal	007
TT-GF-07	AI	GF-07 Rest room	Required	007
RHT-GF-07	AI	GF-07 Rest room	Required	007
DPT-GF-07	AI	GF-07 Rest room	Proposal	007
TSHH-EH-07	DI	EH-07	OEM safety	014
TT-GF-03	AI	GF-03 Kitchen	Required	007
RHT-GF-03	AI	GF-03 Kitchen	Required	007
DPT-GF-03	AI	GF-03 Kitchen	Proposal	007
TSHH-EH-03	DI	EH-03	OEM safety	014
TT-GF-01	AI	GF-01 Clean agent	Required	007
RHT-GF-01	AI	GF-01 Clean agent	Proposal	007
DPT-GF-01	AI	GF-01 Clean agent	Proposal	007
TT-SA-01	AI	Common supply	Required	007 / 008 / 012
TT-RA-01	AI	Common return	Required	007 / 012
RHT-RA-01	AI	Common return	Required	007 / 012
FIT-SA-01	AI	Common SA	Required	007 / 008 / 012
FIT-RA-01	AI	Common RA	Required	007 / 012
PDT-OA-01	AI	F-OA-01	Proposal	012
PDI-OA-01	LOCAL	F-OA-01	Proposal	012
TSHH-EH-COM-01	DI	EH-COM-01	OEM safety	014
PDT-F1-A	AI	PAU-A filter 1	Required	012
PDI-F1-A	LOCAL	PAU-A filter 1	Required	012
PDT-F2-A	AI	PAU-A filter 2	Required	012

85 unique instruments; each has a tagged graphic and drawing reference. Proposal / OEM safety / conditional metering statuses are retained. D-sheet references are ADM-HVAC-D-nnn; local PDI instruments are not wired AI points.

Instrument tag	I/O	Service / location	Status	D-sheet
PDI-F2-A	LOCAL	PAU-A filter 2	Required	012
DPT-SF-A	AI	PAU-A fan	Required	012
VS-PAU-A	AI	PAU-A frame / fan bearing	Required	012
PDT-F1-B	AI	PAU-B filter 1	Required	012
PDI-F1-B	LOCAL	PAU-B filter 1	Required	012
PDT-F2-B	AI	PAU-B filter 2	Required	012
PDI-F2-B	LOCAL	PAU-B filter 2	Required	012
DPT-SF-B	AI	PAU-B fan	Required	012
VS-PAU-B	AI	PAU-B frame / fan bearing	Required	012
PDT-F1-C	AI	PAU-C filter 1	Required	012
PDI-F1-C	LOCAL	PAU-C filter 1	Required	012
PDT-F2-C	AI	PAU-C filter 2	Required	012
PDI-F2-C	LOCAL	PAU-C filter 2	Required	012
DPT-SF-C	AI	PAU-C fan	Required	012
VS-PAU-C	AI	PAU-C frame / fan bearing	Required	012
DPT-EF-03-01	AI	EF-03-01	Proposal	013
VS-EF-03-01	AI	EF-03-01	Required	013
DPT-EF-03-02	AI	EF-03-02	Proposal	013
VS-EF-03-02	AI	EF-03-02	Required	013
DPT-EF-09-01	AI	EF-09-01	Proposal	013
VS-EF-09-01	AI	EF-09-01	Required	013
DPT-EF-09-02	AI	EF-09-02	Proposal	013
VS-EF-09-02	AI	EF-09-02	Required	013
DPT-EF-01-01	AI	EF-01-01	Proposal	013
VS-EF-01-01	AI	EF-01-01	Required	013
DPT-SF-B1-01	AI	SF-B1-01	Proposal	013
VS-SF-B1-01	AI	SF-B1-01	Required	013
DPT-SF-B1-02	AI	SF-B1-02	Proposal	013
VS-SF-B1-02	AI	SF-B1-02	Required	013
DPT-EF-B1-01	AI	EF-B1-01	Proposal	013
VS-EF-B1-01	AI	EF-B1-01	Required	013
DPT-EF-B1-02	AI	EF-B1-02	Proposal	013
VS-EF-B1-02	AI	EF-B1-02	Required	013
TT-B1-01	AI	Cable basement	Required	015
DPT-B1-01	AI	Cable basement	Proposal	015
EM-HPCP-01	COMM	HPCP feeders >7.5 kW	Required*	015
AFS-EH-04	DI	Electric heater EH-04	Proposal	014
AFS-EH-05	DI	Electric heater EH-05	Proposal	014
AFS-EH-06	DI	Electric heater EH-06	Proposal	014
AFS-EH-07	DI	Electric heater EH-07	Proposal	014
AFS-EH-03	DI	Electric heater EH-03	Proposal	014
AFS-EH-COM-01	DI	Electric heater EH-COM-01	Proposal	014

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-016

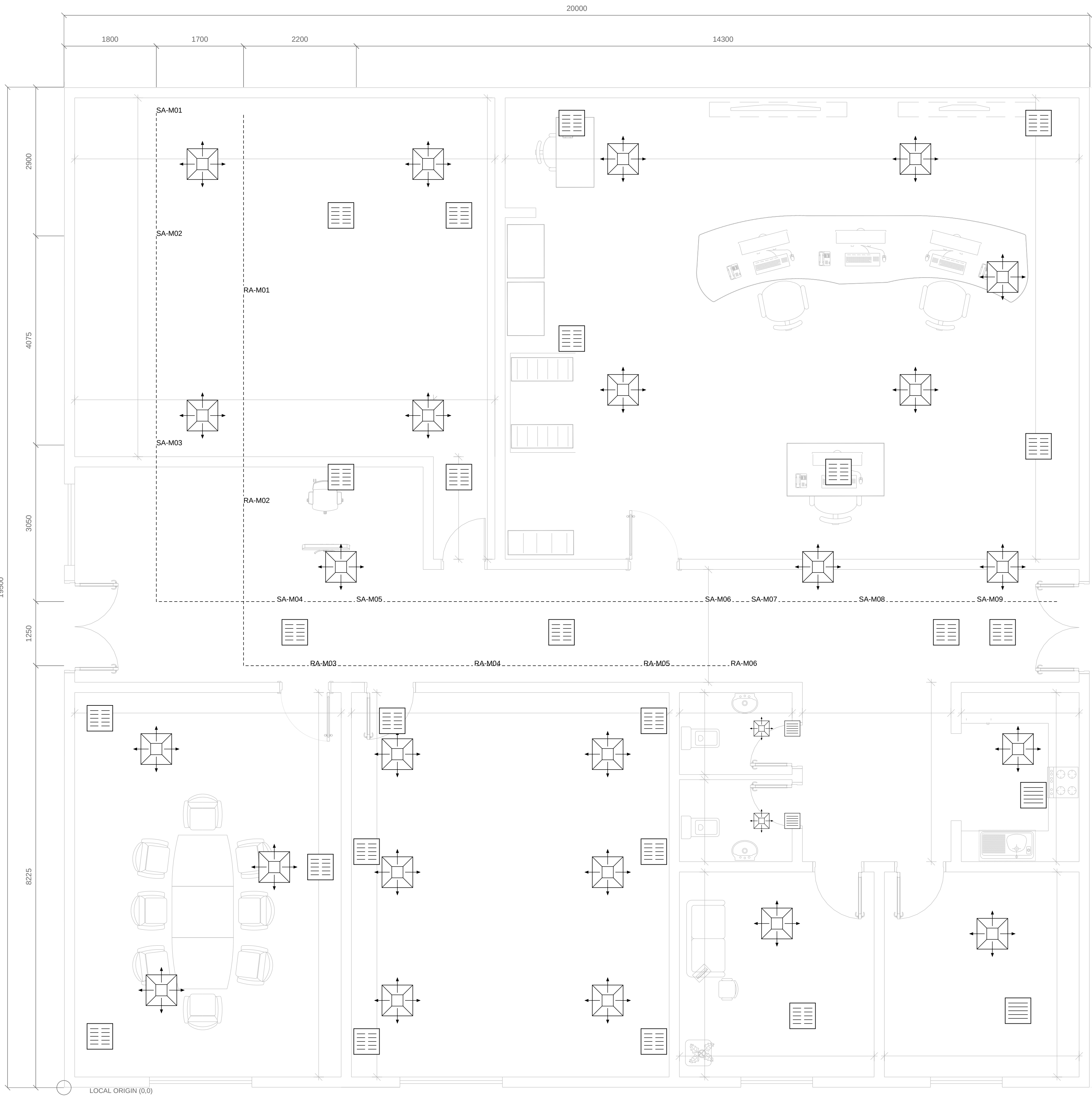
SCALE: NTS AT A1

REVISION

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Dimension chains are plan centerline offsets and architectural extents only. They are not approved fabrication, support or vertical-clearance dimensions. Terminal faces retain scheduled dimensions; final catalogue free areas and performance remain pending.

SOURCE PLAN COORDINATES / DATUM

Dimensions and coordinates are derived from the supplied architectural plan. Local (0,0) is a drawing origin, not a surveyed site benchmark. Confirm all setting-out against the coordinated ceiling and architecture.

GF	Terminal X range mm	Terminal Y range mm
02	2700 - 7700	11900 - 18000
05	9900 - 19000	12000 - 18800
04	700 - 5000	1000 - 7200
08	5900 - 11500	900 - 7150
06	4500 - 18300	8875 - 10150
07	13900 - 14400	1400 - 3200
09	13600 - 14200	5200 - 7000
03	18600 - 18900	5700 - 6600
01	18100 - 18600	1500 - 3000

Individual terminal coordinates: Terminals register. Main centerlines: Duct_sections register. Native CAD dimensions are provided for the known overall extents and main offsets.

Structural underside is +4.000 m (user confirmed). External insulation 50 mm is an indoor coordination assumption. Duct BOD and finished ceiling remain TBC. D-020 records the crossing stack limits and CX-08 height hold.

INSPECTION / COORDINATION NOTES

- Verify dimensions on site; do not scale NTS details.
- Match device tags to registers before procurement.
- Reserve access to dampers, heaters and filters.
- Obtain selected-fitting losses and OEM duties.
- Close pressure / outdoor-air design holds.

ADMINISTRATION BUILDING - DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

DETAILED DRAWING INDEX

Drawing number	Drawing title	Scale
ADM-HVAC-D-001	COORDINATED DUCT PLAN	1:50
ADM-HVAC-D-002	SUPPLY AIR DUCT PLAN	1:50
ADM-HVAC-D-003	RETURN AIR AND EXTRACT DUCT PLAN	1:50
ADM-HVAC-D-004	NORTH ROOMS - DUCT ENLARGEMENT	1:35
ADM-HVAC-D-005	SOUTH-WEST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-006	SOUTH-EAST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-007	ROOM INSTRUMENTATION AND CONTROL LOCATIONS	1:50
ADM-HVAC-D-008	PAU BANK AND INDOOR RISER SECTION	NTS
ADM-HVAC-D-009	DUCT SYMBOLS AND INSTALLATION DETAILS	NTS
ADM-HVAC-D-010	DUCT AND TERMINAL SCHEDULES	NTS
ADM-HVAC-D-011	TAPERED FITTINGS AND MAINTENANCE ACCESS	NTS
ADM-HVAC-D-012	PLANT INSTRUMENTATION AND HEATER PROTECTION	NTS
ADM-HVAC-D-013	EXTRACT AND BASEMENT FAN INSTRUMENTATION	NTS
ADM-HVAC-D-014	ELECTRIC HEATER PROTECTION AND CONTROL	NTS
ADM-HVAC-D-015	FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES	NTS
ADM-HVAC-D-016	COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES	NTS
ADM-HVAC-D-017	PLAN DIMENSIONS AND TERMINAL SETTING-OUT	1:50
ADM-HVAC-D-018	DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES	NTS
ADM-HVAC-D-019	ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS	AS SHOWN
ADM-HVAC-D-020	CROSSING REGISTER / VERTICAL COORDINATION	NTS
ADM-HVAC-D-021	REV10 TERMINAL / SIZE CHANGES	NTS

DESIGN BASIS / OPEN ENGINEERING ITEMS

Supply 3697.80 L/s; assumed return 3006.68 L/s. Pressure targets +50 Pa normal rooms, +25 Pa kitchen / toilets / clean-agent; actual balance unverified.

OA reference 420.70 L/s is below 691.12 L/s implied makeup. Toilet net SA-EA is -18.69 L/s; kitchen net is zero. Neither proves the +25 Pa target. Resolve transfer air / extract balance and outdoor-air compliance.

Fan ESP / curves, fitting K, vertical lengths, levels, support and insulation, diffuser throw / noise, access envelopes and I&C settings remain pending.

ISSUE STATUS: FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION. Drawn / checked / approved identities and authority are pending. Reference styling does not establish design compliance or professional sign-off.

REVISION 10 - SCOPE OF ISSUE

Routing revision: remove same-service crossovers / duplicate runouts, separate coincident service tracks and render connected exterior walls once.

Complete tagged fan / heater protection and basement / energy interfaces are added. Existing 85 instrument requirements and room duties are retained. Proposed section size changes and terminal relocations are scheduled on D-021.

Reference D-113391-ADDC-AES-ME-AC-001_REV.1 is a drafting benchmark; its building duties and instrument requirements are not substituted for this Administration design.

DRAWING / COORDINATION CONVENTIONS

All duct sizes: clear internal width x height in mm; D denotes round diameter. Flows: L/s; velocities: m/s. Add approved insulation outside clear sizes.

Numbered CX tags identify separate-service crossings. Dashed lower edges / continuous upper edges refer to D-020; proposed levels and clearance remain pending.

MFD* graphic corresponds to conditional FSD-prefixed register tags. Approval of rated boundaries governs quantity, rating and fail positions.

TT / RHT / DPT room locations are proposed. Pressure references are outdoors. Dashed I&C lines are functional signals, not installed cable routes.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-018

SCALE: NTS AT A1

REVISION

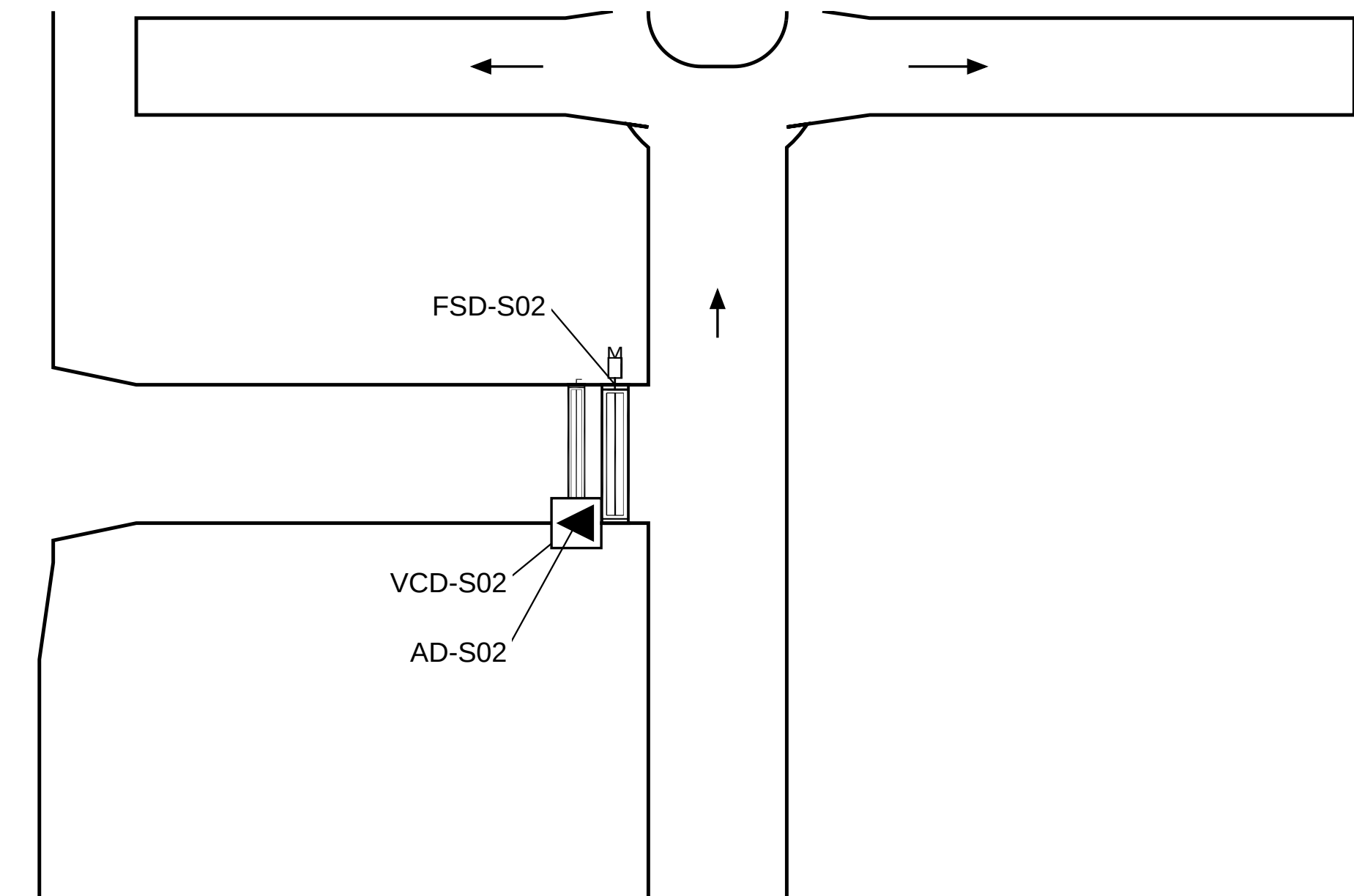
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ADMINISTRATION BUILDING - ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

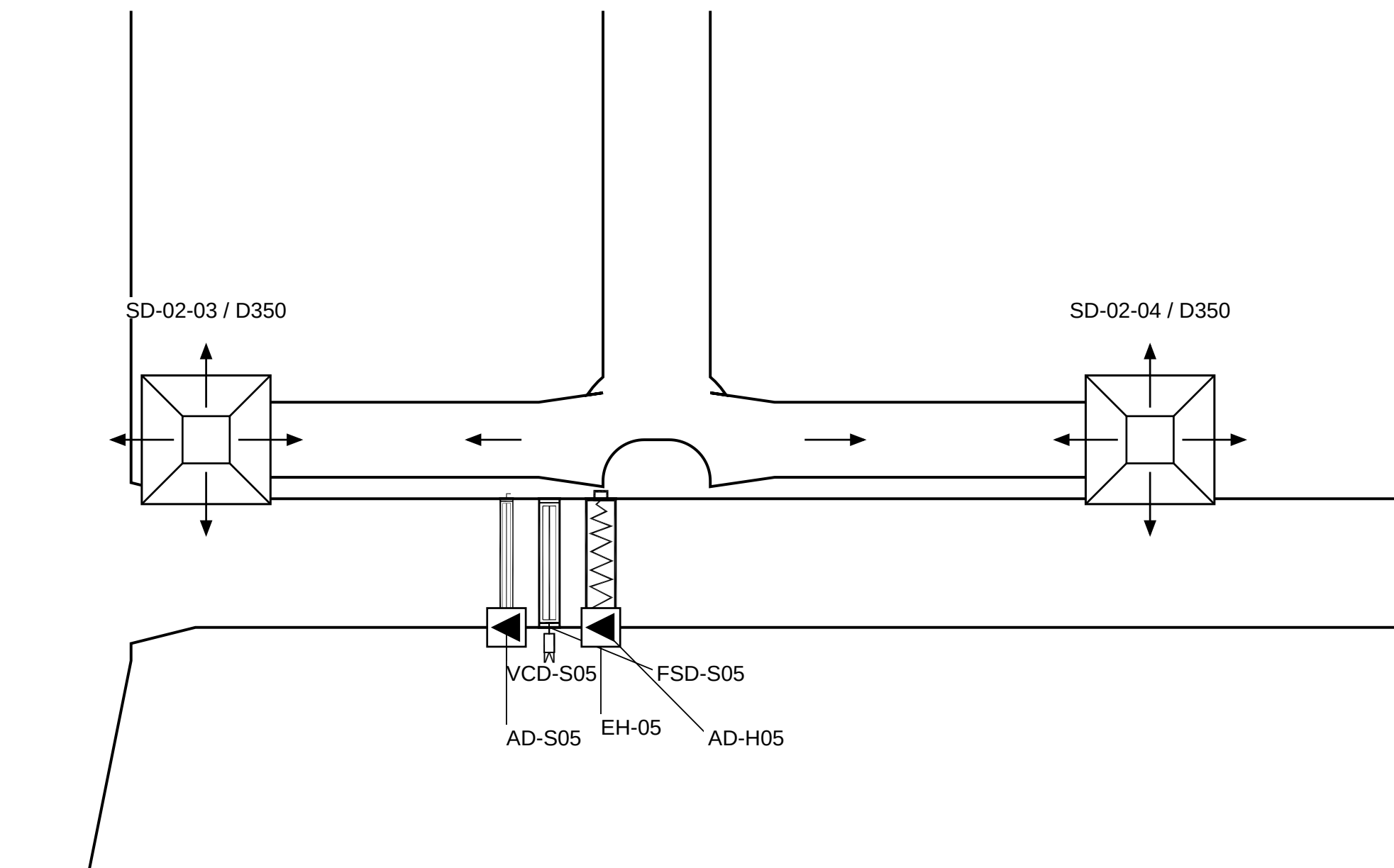
A GF-02 BRANCH / TWO HEADER LEGS / 1:20



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

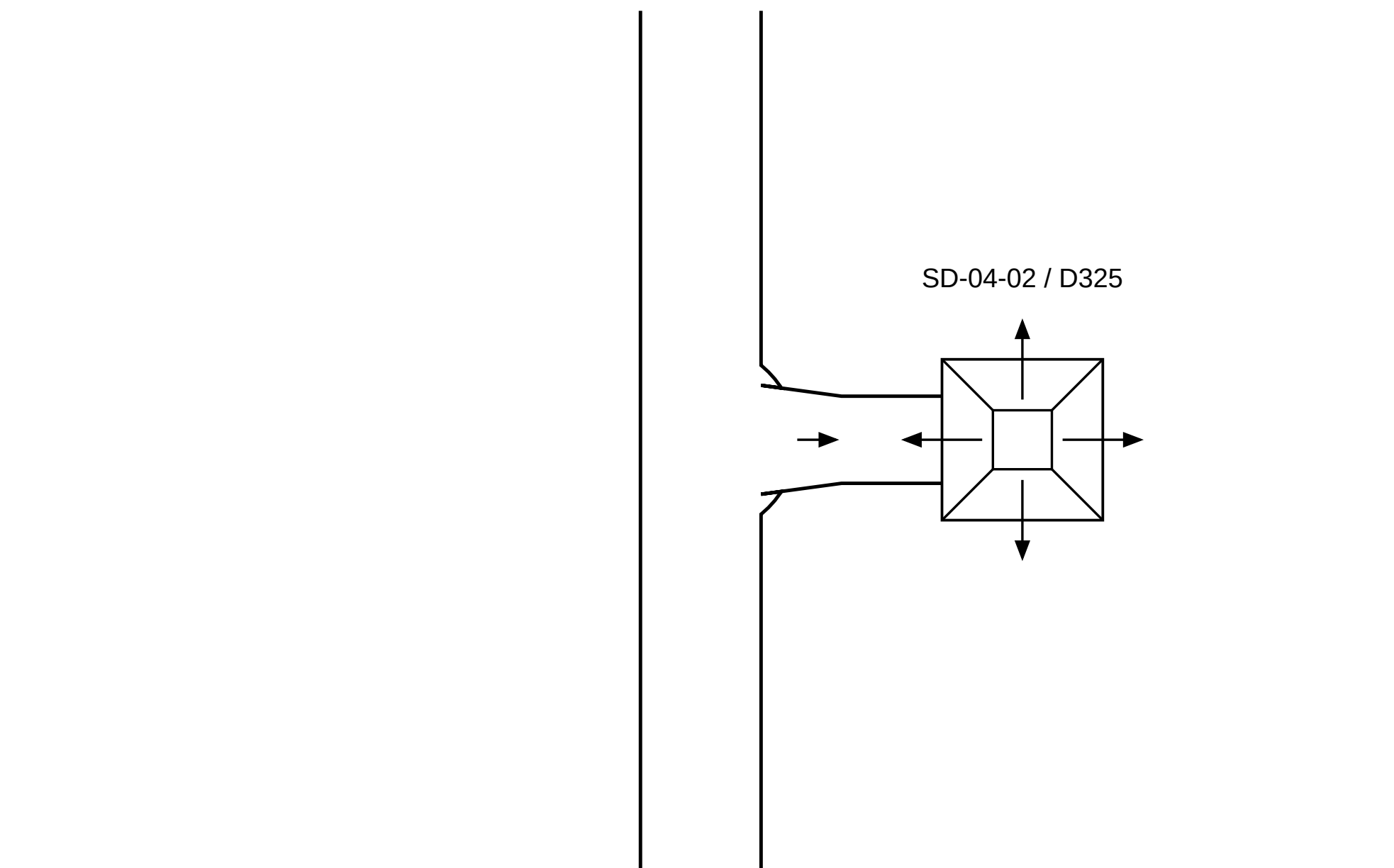
B GF-02 END TEE / TWO INDIVIDUAL NECKS / 1:25



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

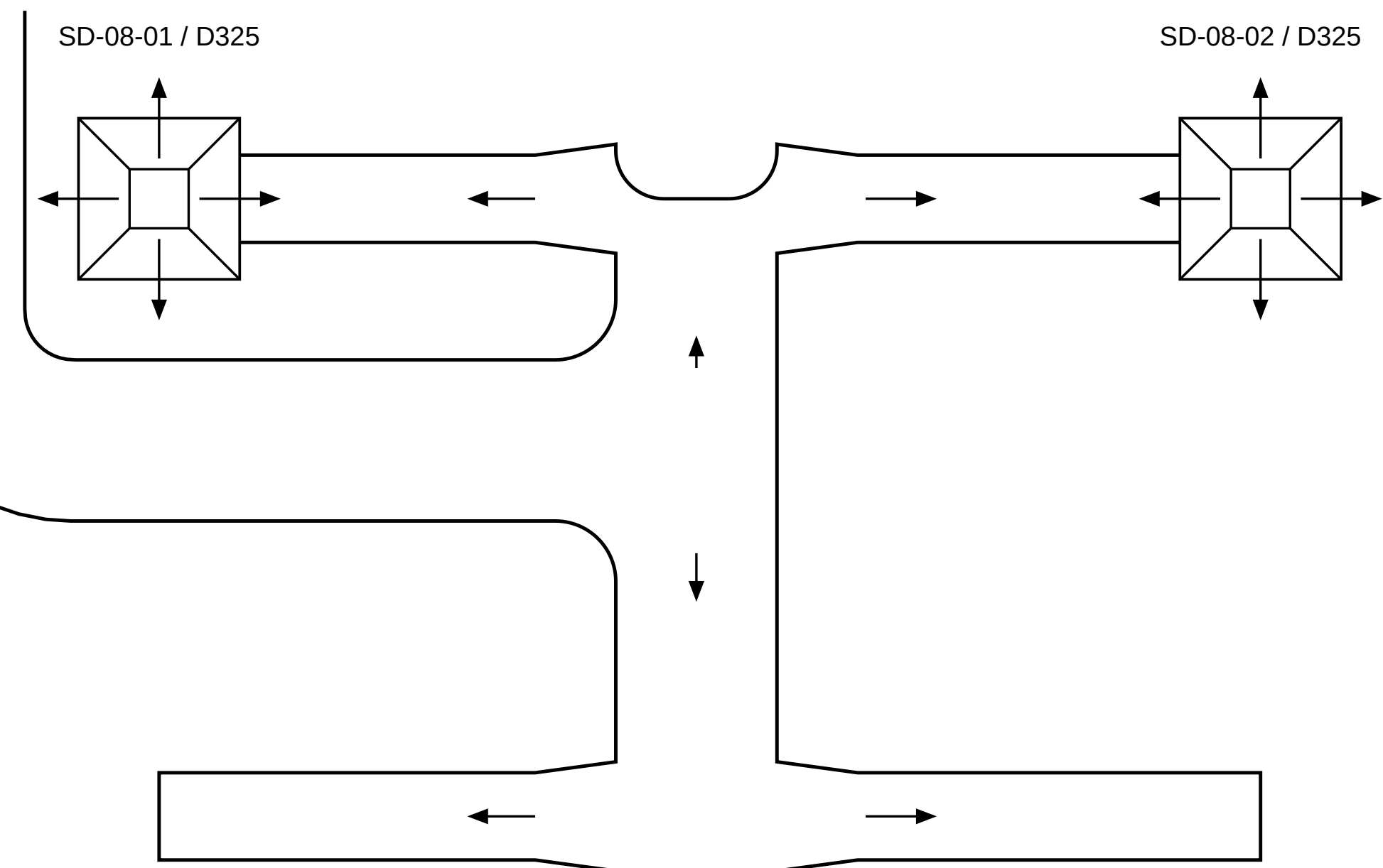
C MEETING CONTINUING HEADER / SIDE NECK / 1:20



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

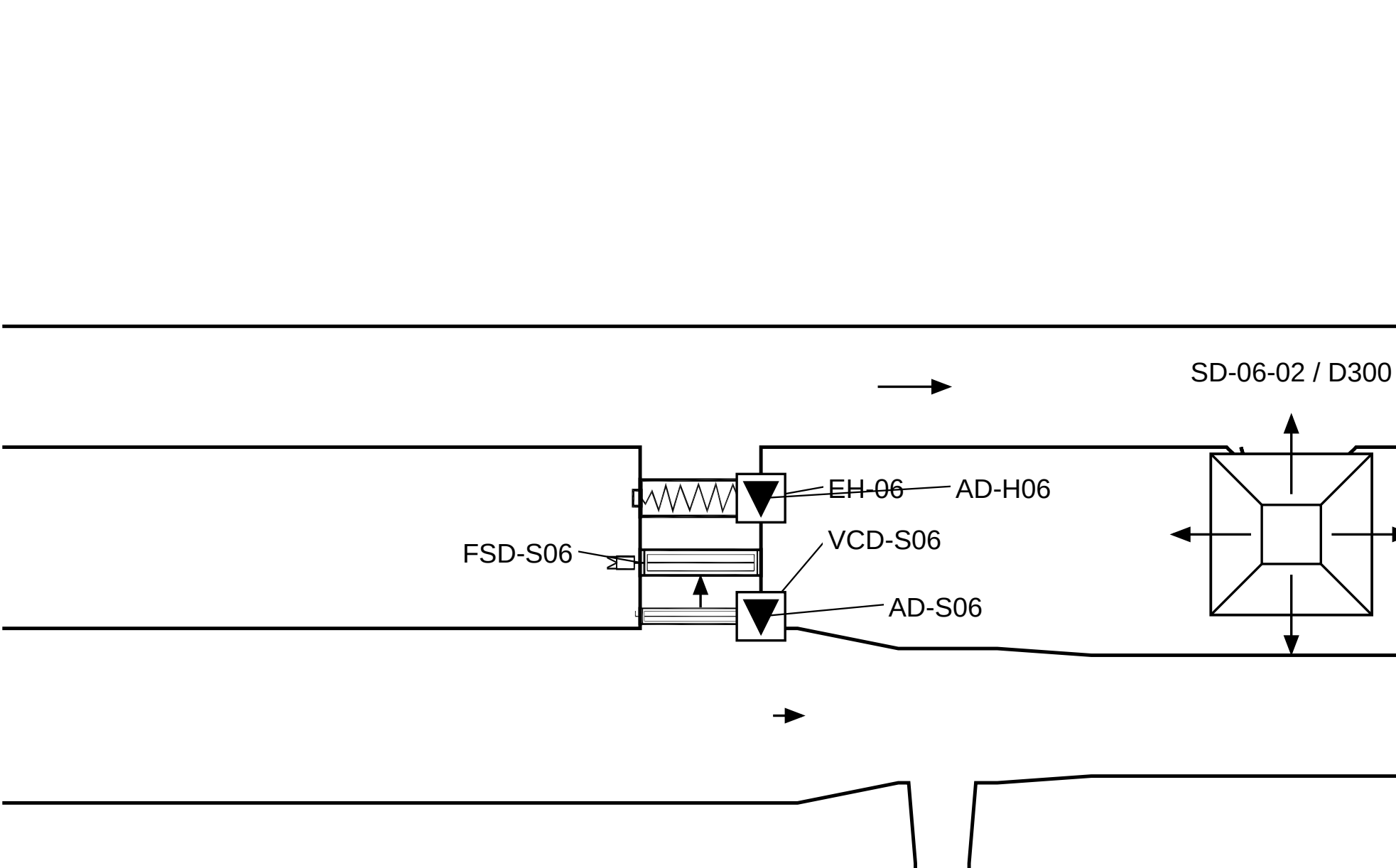
D TELECOM FEEDER / UP AND DOWN HEADERS / 1:20



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

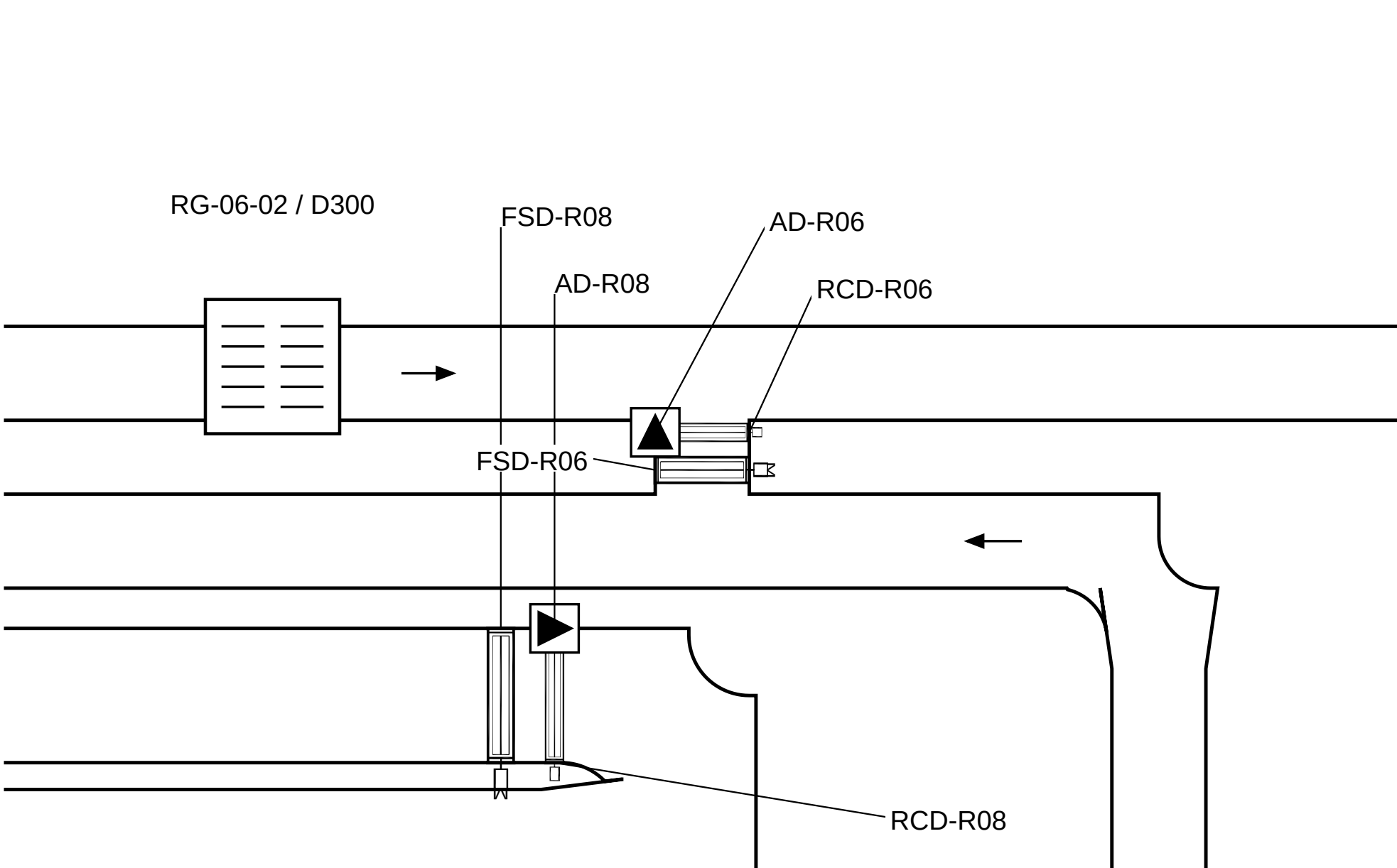
E CORRIDOR COMMON BRANCH / HEATER / HEADER / 1:20



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

F CORRIDOR COLLECTION / RETURN DEVICE STATION / 1:20



Actual Rev10 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; coordinate CX references on D-020.

These details are generated from the delivered routing geometry, not independent typical symbols. Continuous exteriors, flared necks and concave fitting throats are checked. Short necks use an integral plenum / vertical drop; their selected construction is pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-019

SCALE: AS SHOWN AT A1

REVISION

10

07 OCT 2026

SHEET 19 OF 21

ADMINISTRATION BUILDING - DUCT CROSSING REGISTER AND VERTICAL COORDINATION

NUMBERED SEPARATE-SERVICE CROSSINGS / D-001 TO D-006

CX	Upper / lower*	Upper duct sections	Lower duct sections	X, Y mm
CX-01	SA / RA	SA-R02-01	RA-M01	3500, 18000
CX-02	SA / RA	SA-B02	RA-M01 / RA-M02	3500, 16652.5
CX-03	SA / RA	SA-H02-D01	RA-B02	4800, 15500
CX-04	SA / RA	SA-R02-03	RA-M03	3500, 13100
CX-05	SA / RA	SA-H02-D04	RA-H02-D01	6000, 13100
CX-06	SA / RA	SA-B05	RA-M03	3500, 12525
CX-07	SA / RA	SA-B05	RA-H02-D01	6000, 12525
CX-08	SA / RA	SA-M03	RA-M03	3500, 9475
CX-09	SA / RA	SA-B03	RA-H06-E01	17800, 8825
CX-10	SA / RA	SA-B07	RA-H06-D01	15500, 8812.5
CX-11	SA / RA	SA-B09	RA-H06-D01	13400, 8800
CX-12	SA / RA	SA-B08	RA-H06-L01	5700, 8762.5
CX-13	SA / RA	SA-B04	RA-M03 / RA-M04	4150, 8225
CX-14	SA / RA	SA-B08	RA-M04	5700, 8225
CX-15	SA / EA	SA-B03	EA-B09	17800, 7630.7
CX-16	SA / EA	SA-B07	EA-B09	15500, 7600
CX-17	SA / EA	SA-B01	EA-B09	19400, 7600
CX-18	SA / RA	SA-B04	RA-R04-01	4150, 7200
CX-19	SA / EA	SA-B01	EA-B03	19400, 5700
CX-20	SA / RA	SA-H08-D01	RA-H08-03	8500, 4600

* Proposed vertical order, not verified installed levels. Every crossing requires BOD, ceiling / structure, insulation and support clearance to be coordinated. No same-service intersection is concealed by this convention.

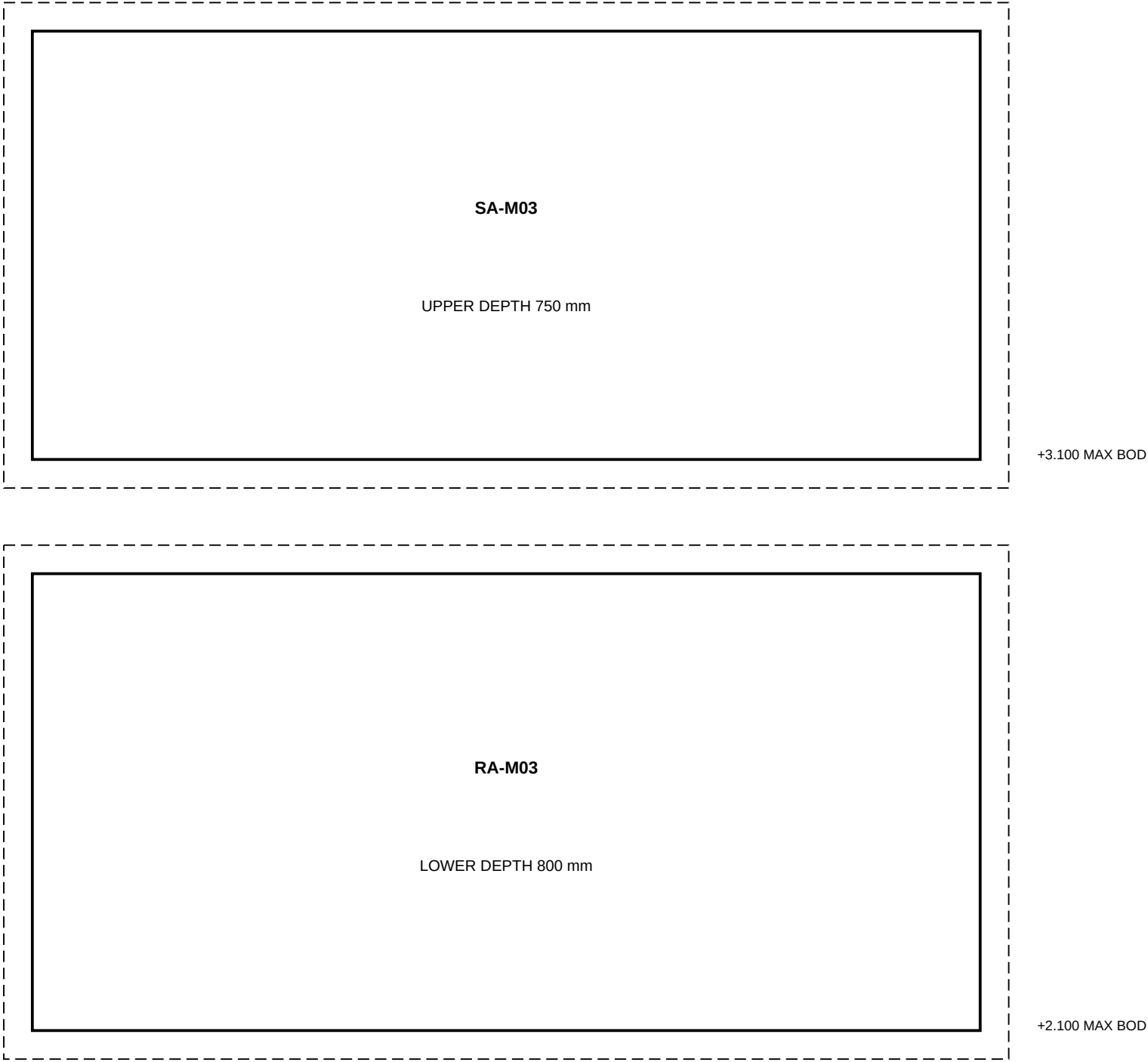
Dashed lower-service edges are limited to the actual overlap footprint. The upper duct remains continuous. CX-01 through CX-20 are searchable in CAD, PDF and Excel.

Structural underside: +4.000 m, user confirmed. Insulation: 50 mm indoor coordination assumption. Every crossing register includes the required stack and maximum local BOD bounds; these are not assigned duct elevations.

COORDINATION HOLD: 4.000 m structural underside confirmed; finished ceiling and duct BOD remain unassigned. Maximum local elevation bounds are calculated, not approved. Check the full crossing register and resolve CX-08 before construction.

CX-08 HEIGHT ENVELOPE - NTS / HOLD

STRUCTURAL UNDERSIDE +4.000 m (USER)



ASSUMED: 50 mm external insulation each duct; 100 mm gap between insulation; 100 mm structure / hanger allowance.

Required stack: 1950 mm. Lowest insulated surface: +2.050 m under those assumptions. Check finished ceiling and access before selecting any level.

CX-08 needs a shallower crossing section or reroute if the finished ceiling / access allowance cannot fit below this envelope. Recalculate width, velocity, transitions and fitting losses for the selected solution.

All BODs remain TBC. Local bounds are not a connected-network elevation design. Insulation is an explicit assumption, not a verified UAE standard. Actual beams, supports, OEM clearances and terminal plenums are not surveyed.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DUCT CROSSING REGISTER AND VERTICAL COORDINATION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-020

SCALE: NTS AT A1

REVISION

10

07 OCT 2026

SHEET 20 OF 21

ADMINISTRATION BUILDING - REV10 TERMINAL MOVES AND DUCT SIZE CHANGES

PROPOSED TERMINAL RELOCATIONS - ROOM DUTIES UNCHANGED

Terminal	Old X, Y mm	New X, Y mm	L/s
RG-02-01	1200, 17000	5400, 17000	151.37
RG-02-03	1200, 11900	5400, 11900	151.37
SD-04-02	4300, 4300	4100, 4300	149.73
RG-04-01	700, 7000	700, 7200	129.95
RG-04-02	4900, 4300	5000, 4300	129.95
RG-08-01	5900, 7100	6400, 7150	137.54
RG-08-02	11200, 7100	11500, 7150	137.54
RG-08-04	11200, 4600	11500, 4600	137.54
RG-08-06	11200, 900	11500, 900	137.54
RG-05-01	9700, 18800	9900, 18800	143.67
RG-05-03	9700, 14600	9900, 14600	143.67
RG-05-05	15100, 10700	15100, 12000	143.67
SD-06-01	5400, 9800	5400, 10150	139.40
SD-06-02	14700, 9800	14700, 10150	139.40
SD-06-03	16300, 6000	16300, 10150	139.40
RG-06-01	3100, 8550	4500, 8875	89.72
RG-06-02	9700, 8550	9700, 8875	89.72
RG-06-03	17200, 8550	17200, 8875	89.72
RG-06-04	15100, 5600	16300, 8875	89.72
SD-09-01	13200, 7000	13600, 7000	23.10
SD-09-02	13200, 5200	13600, 5200	23.10
SD-01-01	18100, 3200	18100, 3000	141.30

The revised points are proposals. Coordinate all grille / diffuser faces against the reflected ceiling plan, lighting, access, throws and noise. Corridor terminals are returned to the corridor arrangement. Individual air duties, neck sizes and instrument allocations remain unchanged.

PROPOSED CLEAR DUCT SIZE REVISIONS

Section	Rev09 mm	Rev10 mm
SA-M03	950x500	650x750
SA-M04	850x425	650x575
RA-M03	850x550	600x800
RA-M04	800x450	600x600
RA-M05	650x300	350x575
RA-M06	450x225	350x300
SA-B04	450x300	350x400
RA-B08	700x450	500x650
RA-H08-U01	700x450	500x650
RA-H08-D01	700x450	500x650
RA-H08-D02	700x450	500x650
RA-B05	650x400	500x525
RA-H05-U01	650x400	500x525
RA-H05-U02	650x400	500x525
RA-H05-U03	650x400	500x525
RA-H05-D01	650x400	500x525
RA-B06	550x350	350x550
RA-H06-D01	550x350	350x550

Narrower / deeper sections separate the plan tracks. Areas and velocities are recalculated; the largest proposed depth is 800 mm. Available ceiling space, both height and width transitions, selected loss coefficients and fan ESP require verification.

ROUTING AND GEOMETRY CHECKS

122 scheduled section paths; every joint conserves its registered airflow.

No duplicated same-service centerline and no unintended same-service polygon intersection.

Separate main / corridor collection tracks; shared runs are sized headers, not coincident individual runouts.

Continuous service exteriors; tapers replace the original throat; local formed fitting radii are shown.

20 separate-service crossings are indexed on D-020; elevations remain pending.

85 instrument tags and the existing I/O duties are retained.